

Global Spillovers from U.S. Bond Supply Shocks*

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Abstract

I study spillovers from U.S. bond supply shocks, identified from yield moves around U.S. Treasury auction announcements. Using the shocks as instruments in daily and monthly local projections for AEs and EMs, I find pass through to global long rates. A debt expansion shock that raises the U.S. 10 year yield by 10 bps lifts foreign 10 year yields by roughly 7 bps within a few business days. Monthly aggregation shows contractionary real effects: U.S. industrial production falls up to 0.4 percent after eight months and foreign industrial production by a comparable amount, around 0.4 percent, with a more persistent profile. The foreign contraction reflects financial tightening, exchange rate depreciation, and weaker U.S. demand. Spillovers are heterogeneous: moving from the 25th to 75th percentile of foreign investor participation deepens IP decline by 0.5 pp. Results are robust to a demand side shock and consistent with benchmarks, highlighting U.S. bond supply as a key driver of global financial conditions and real activity, amplified by foreign investor participation and weak domestic fundamentals.

JEL codes: E44, E62, F42, G15.

Keywords: fiscal spillovers; bond supply shocks; local projections; high-frequency identification; international transmission.

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1 Introduction

Government bond markets are undergoing major structural shifts in both supply and demand. As public debt levels in advanced economies have risen sharply since the global financial crisis and the COVID-19 pandemic, sovereign bond issuance has expanded in size and, in many cases, shortened in duration. These shifts raise the sensitivity of bond prices to issuance news and, through integrated international capital markets, transmit to asset prices and real economic conditions well beyond the country of issuance. Understanding how U.S. Treasury supply shocks propagate globally is therefore central to assessing the cross-border consequences of U.S. fiscal policy.

This paper quantifies the international spillovers of unexpected changes in U.S. Treasury issuance. I take as given the high-frequency U.S. Treasury supply shocks constructed by Phillot (2025), who exploits narrow 30-minute windows around scheduled Treasury auction announcements to isolate within-window yield changes that reflect pure supply surprises, stripped of concurrent macroeconomic news and Federal Reserve communications. Following the structural decomposition of Bi et al. (2026), I apply principal component analysis to the cross-maturity pattern of within-window yield changes and extract two shocks: a *debt expansion shock*, capturing unanticipated increases in the total stock of outstanding debt that broadly lift the yield curve; and a *maturity extension shock*, capturing compositional surprises that steepen the yield curve without altering its overall level. These shocks and their domestic macroeconomic effects have been studied by Phillot (2025) and Bi et al. (2026); the contribution of this paper is to characterize their international transmission to foreign sovereign bond markets and to real activity and the country characteristics that govern how large those spillovers are. The publicly available shocks of Phillot (2025) end in January 2020; I extend them through February 2026 using surprises I construct myself from intra-day Bloomberg tick data under the identical identification, which adds coverage of the high-issuance post-COVID period.

My empirical strategy proceeds in two steps. In the first step, I estimate daily local projections (LPs) to trace the dynamic response of U.S. and foreign 10-year sovereign yields to these shocks over horizons up to ten business days, for a panel of up to 39 economies. In the second step, I aggregate the daily shocks to monthly frequency following the weighting scheme of Gertler and Karadi (2015) and estimate monthly LPs to measure the effects on industrial production (IP) domestically and internationally, over horizons of up to 24 months, across a panel of up to 58 economies. Throughout both steps, I evaluate heterogeneity by interacting the shocks with country characteristics that capture financial-market exposure and macroeconomic resilience, exploiting variation across countries within time periods using time fixed effects.

My main results are as follows. First, debt expansion shocks that raise the U.S. 10-year yield by 10 basis points transmit strongly to foreign sovereign bond markets, lifting a broad panel of foreign 10-year yields by approximately 7 basis points within three business days. When I normalize the foreign yield response by the shock-induced U.S. yield movement using an instrumental variables (IV) approach, the implied spillover elasticity reaches approximately 0.7 within three days, meaning that foreign bond markets absorb roughly 70 percent of the U.S. yield increase induced by the supply

shock. The first-stage Kleibergen-Paap F-statistic of 26 confirms instrument strength.

Second, at monthly frequency the debt expansion shock has a contractionary and persistent effect on U.S. macroeconomic conditions, consistent with a tightening in financial conditions. The U.S. 10-year yield rises, CPI falls, and industrial production declines by up to 0.4 percent at an eight-month horizon. These domestic results closely replicate the benchmark estimates of Bi et al. (2026), providing external validation for my shock series and specification.

Third, the contractionary effects on real activity extend abroad, with a foreign industrial production decline of comparable magnitude to the U.S. impact (around 0.4 percent) but a more persistent profile. Using exposure-based channel designs, I provide suggestive evidence consistent with three mechanisms: a financial channel, whereby shock-induced increases in U.S. yields tighten global financial conditions and raise borrowing costs for foreign firms and sovereigns; exchange rate channel, whereby foreign economies experience currency depreciation; and a trade channel, whereby the reduction in U.S. activity reduces demand for foreign exports. The financial and trade conduits both contribute contractionary spillovers of the expected sign, while the exchange-rate conduit is more weakly identified. I additionally show that inflation falls in foreign economies, consistent with the overall contraction.

Fourth, spillovers are substantially heterogeneous across countries. I organize the cross-country evidence around two dimensions. First, financial market integration: economies with a larger share of sovereign bonds held by foreign investors or with more open capital accounts experience industrial production contractions roughly 0.5 percentage points deeper at medium horizons (IQR-scaled difference between 75th and 25th percentiles). Second, macroeconomic fundamentals, which I separate into macro strength and borrowing capacity. On the macro strength side, countries with larger fiscal space and stronger current account positions are more resilient: they have the policy room and external financing independence to absorb the shock, exhibiting significantly smaller output contractions and lower yield spillovers. On the borrowing capacity side, countries with more volatile sovereign debt markets or greater debt-at-risk face amplified spillovers: the global repricing of safe assets disproportionately tightens their market access, generating outsized tightening effects on borrowing costs and real activity.

Finally, as a robustness check and to contextualize the magnitude of supply-side effects, I compare my results against a demand-side bond market shock: the large-scale asset purchase (LSAP) component of U.S. monetary policy surprises from Swanson (2021). The estimated spillover transmission profiles are broadly comparable in shape, suggesting that my supply-shock findings reflect a systematic feature of U.S. bond market transmission rather than an artifact of identification.

Related literature. This paper contributes to several strands of the literature. First, it adds to work on the international spillovers of U.S. monetary and fiscal policy. A large body of research documents that U.S. interest rate and monetary policy shocks transmit to foreign asset prices and real activity through the global financial cycle (e.g., Rey, 2015; Miranda-Agrippino and Rey, 2020; Carrière-Swallow et al., 2025). A notable feature of this literature is that spillovers from

the monetary policy of the U.S. and other advanced economies, such as the euro area and Japan, are large and systematic, reaching advanced as well as emerging markets. I complement this work by shifting attention to the *supply* side of the U.S. bond market, and documenting that Treasury issuance shocks constitute an independent and quantitatively important channel of global transmission, distinct from conventional monetary policy shocks.

Second, the paper builds directly on the literature on high-frequency identification of fiscal and debt supply shocks, and on the role of foreign investors in sovereign bond markets (Arslanalp and Tsuda, 2014a,b). The identification and domestic analysis are not my contribution: Phillot (2025) constructs the raw auction-based shock series I use, and Bi et al. (2026) introduce the structural decomposition I adopt and establish the domestic U.S. macroeconomic effects, which I take as given and reproduce only as a validation check. My contribution is to take these shocks to the international dimension—estimating spillovers to a panel of over 50 economies, decomposing the transmission channels, and characterizing the country characteristics that govern their size—and to extend the shock series itself beyond its publicly available 2020 endpoint, constructing post-2020 surprises from intraday Bloomberg tick data under the identical identification so as to study spillovers in the high-issuance post-COVID period (Section 4.4, Appendix A). The finding that foreign investor participation amplifies both yield and output spillovers connects to the broader portfolio-balance literature on how cross-border holdings shape the international propagation of financial shocks. It relates particularly closely to Arcidiacono et al. (2024), who identify an analogous debt-supply spillover among euro-area sovereigns—near one-for-one transmission of convenience-yield shocks between low-risk countries and weaker transmission to riskier ones—and rationalize it with a model in which equally safe bonds are close substitutes. My results extend this substitutability logic to the global transmission of *U.S.* supply shocks and document the same risk-dependent gradient in the cross-country heterogeneity (Section 4.9).

Third, the paper connects to a substantial literature studying the effect of the level and composition of government debt on long-term interest rates and term premia. Seminal contributions by Laubach (2009) and Engen and Hubbard (2004) estimate that a one-percentage-point increase in the projected debt-to-GDP ratio raises long-term real interest rates by approximately 3–4 basis points and 2–3 basis points, respectively. The CBO’s analysis, building on this framework, reports estimates of a similar order (Congressional Budget Office, 2019). Cross-country evidence from Warnock and Warnock (2009) and Kitchen and Chinn (2011) further documents the role of foreign holdings in moderating or amplifying these effects. My high-frequency identification approach complements this literature: while I cannot directly observe the size of the underlying issuance surprise in debt-to-GDP terms, the daily yield impact of approximately 1–2 basis points per one-standard-deviation shock is broadly consistent with the semi-elasticities estimated in the debt-interest-rate literature, assuming the auction surprise corresponds to a modest revision in the expected debt trajectory on the order of 0.3–0.5 percentage points of GDP.¹

¹The semi-elasticity of 3–4 bp per percentage point of debt-to-GDP from Laubach (2009) would imply that a shock raising the 10-year yield by 1.2 bp corresponds to a revision in the expected debt path of roughly 0.4–0.6 pp of GDP. While the comparison is only approximate—the Laubach estimates use 5-year-ahead projected debt and

The remainder of the paper is organized as follows. Section 2 describes the data. Section 3 presents the empirical methodology. Section 4 reports the results, organized by frequency, channel, and heterogeneity dimension. Section 5 concludes.

2 Data

2.1 U.S. Treasury supply shocks

I use the raw high-frequency U.S. Treasury supply shocks constructed by Phillot (2025) and apply the structural decomposition of Bi et al. (2026) to separate them into two economically interpretable components. The raw shocks are identified from price movements of Treasury futures in a narrow 30-minute window bracketing each scheduled Treasury auction announcement. The identifying assumption, which was established and validated by Phillot (2025), is that within such a window, market participants have no time to receive or process macroeconomic information unrelated to the auction; any observed yield change therefore reflects a reassessment of bond supply conditions rather than a response to fundamentals.

The Treasury auction process and the nature of supply surprises. Understanding the institutional context of Treasury auctions is essential for interpreting the magnitude of my identified shocks. The U.S. Treasury follows a “regular and predictable” issuance framework, a cornerstone of debt management policy that has evolved since the late 1970s (Garbade, 2007). Under this framework, the Treasury maintains a transparent and pre-announced schedule of auction offerings across the maturity spectrum, with the explicit goal of minimizing borrowing costs by reducing supply uncertainty for market participants (Frost, 2024; Bessent, 2025).²

A key feature of this framework is that the Treasury concentrates adjustments to borrowing needs on the short end of the curve, using Treasury bills as a buffer to absorb seasonal and unexpected fluctuations in financing needs, while keeping long-term coupon issuance as stable and predictable as possible (Frost, 2024; Bessent, 2025). Nonetheless, supply surprises do occur for two reasons. First, adjustments to short-term issuance signal future refinancing needs: an unanticipated increase in bill issuance today implies larger rollover requirements in the future, which market participants price into longer-term yields. Second, the minor details of upcoming auctions—exact offering sizes, any adjustments to the quarterly refunding plan, and the allocation across maturities—are announced several days before each auction takes place and can materially affect market pricing, particularly given the massive scale of U.S. Treasury issuance.³ Indeed, the very

forward rates, whereas my shock operates at daily frequency—the orders of magnitude are aligned.

²As Frost (2024) articulates, Treasury’s guiding principle is to “finance the government at the least cost over time” through three interrelated pillars: promoting a broad investor base, issuing on a regular and predictable schedule, and maintaining robust contingency planning for a range of fiscal outcomes. Bessent (2025) reaffirms this framework, noting that bills serve as the “shock absorber” for short-term financing needs while nominal coupon issuance adjusts only gradually.

³As of February 2026 (the end of the extended sample), total marketable Treasury debt outstanding exceeded \$29 trillion, with monthly gross issuance of marketable securities running around \$2.5 trillion across all maturities.

existence of the pre-auction announcement process reflects the market’s need for advance information: if announcements contained no news, they would serve no purpose and could have been eliminated entirely. The fact that the Treasury maintains the announcement process precisely to allow markets time to incorporate new information underscores that these events are informative, even if the surprises are small by design.

Consistent with the Treasury’s emphasis on predictability, the identified supply shocks are small in magnitude. A one-standard-deviation debt expansion shock moves the U.S. 10-year yield by approximately 1.2 basis points on the announcement day. To place this in context, the daily standard deviation of the 10-year Treasury yield over my sample period is approximately 4.7 basis points, so the typical auction surprise accounts for roughly one-third of a normal day’s yield variation. This is by design: the Treasury aims to minimize market disruption, and the “regular and predictable” framework ensures that the bulk of issuance information is already priced in before the announcement window.

The validity of this timing restriction rests on two pillars. First, Phillot (2025) documents that Treasury auction windows do not systematically coincide with scheduled U.S. macroeconomic data releases, Federal Reserve communications, or other high-impact events. Second, as an additional check on exogeneity, I conduct Granger causality tests and find that the debt expansion shock is not predicted by a broad set of daily U.S. and foreign financial and economic indicators—including the U.S. 10-year yield, German and Japanese 10-year yields, U.S. term premiums at the 2-, 5-, and 10-year horizons, breakeven inflation, the VIX, the NASDAQ, the S&P 500, and the oil price—at the 5 percent significance level.

On quantity-based and Quarterly Refunding identification. A natural question is whether the supply surprises could instead be identified from quantity-based measures or restricted to Quarterly Refunding Announcement (QRA/QRS) windows, as in Hartley and Rigon (2025). I rely on the price-based auction-announcement identification of Phillot (2025) for two reasons. First, as Phillot (2025) and Bi et al. (2026) emphasize, realized issuance quantities adjust only gradually under the Treasury’s “regular and predictable” framework and are a noisy, delayed proxy for the forward-looking supply news that markets price; the within-window futures reaction captures the surprise component directly, including the broader issuance-strategy guidance that headline auction sizes miss. Second, the question of whether the QRS-only subset of announcements delivers similar effects has already been settled by the benchmark paper whose decomposition I adopt: Bi et al. (2026) split their shocks into QRS and non-QRS announcements and show that the two sets generate yield responses that move in the same direction and are of similar magnitude (with the non-QRS estimates more precisely estimated, owing to their larger number, and the largest maturity-extension surprises concentrated on QRS dates).

Because auction announcements vary in their maturity composition, within-window yield changes across the four benchmark maturities conflate two conceptually distinct shocks. Following the de-

composition of Bi et al. (2026), I apply principal component analysis to the vector of yield changes across the 2-, 5-, 10-, and 30-year segments.⁴ The two resulting structural shocks are:

- **Debt expansion shock** (s_t^{DE}): the first principal component, loading approximately equally and positively across all four maturities (loadings: +0.48, +0.52, +0.53, +0.47). This shock captures an unanticipated increase in the total outstanding stock of U.S. Treasury debt, which broadly raises the yield curve.⁵
- **Maturity extension shock** (s_t^{ME}): the second principal component, loading with opposing signs at short and long maturities and summing to approximately zero (loadings: −0.78, −0.01, +0.20, +0.59). This shock captures a compositional surprise—an unanticipated shift toward longer-duration issuance—that twists and steepens the yield curve without altering its overall level.

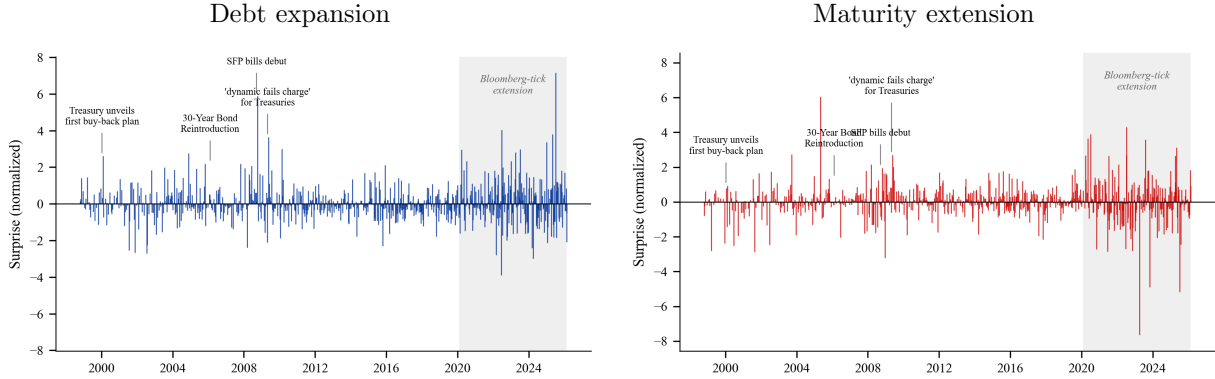
Both series are signed so that a positive realization corresponds to an upward surprise in yields (i.e., a contractionary supply surprise). I extend the shock series through the post-COVID period. The publicly available Phillot (2025) shocks end in January 2020; I splice them with an extension that I construct myself following the identical Phillot (2025) identification (30-minute within-window Treasury futures price changes at the 2-, 5-, 10-, and 30-year benchmark maturities, sign-flipped to express a supply surprise) using intraday Bloomberg tick data, covering February 2020 onward. The merge is described in Appendix A: over the 2013–2020 overlap window the two procedures flag essentially the same auction dates and the maturity-level surprises are strongly positively correlated at each maturity (Pearson 0.66–0.80, OLS slopes near one), I rescale the extension to the published series’ units, and I re-run the Bi et al. (2026) PCA decomposition on the spliced cross-maturity surprise matrix so that the debt-expansion and maturity-extension loadings are estimated on the full sample. The extended sample spans October 1998 to February 2026, yielding a larger set of auction events and, importantly, coverage of the heightened global fiscal uncertainty of the post-COVID period. Figure 1 plots the time series of both shocks; the shocks exhibit little serial correlation or cross-correlation, consistent with their identification as innovations.

Comparison with monetary policy shocks: daily yield variation. It is instructive to compare the daily yield variation introduced by Treasury supply shocks with that induced by the

⁴The main spillover results are robust to bypassing the decomposition entirely and using the raw within-window 10-year surprise directly as the shock: the daily yield spillover, the IV-normalized elasticity, and the monthly IP responses are quantitatively close to the debt-expansion-shock baseline, which is unsurprising given that the debt-expansion factor loads heavily on the 10-year segment and explains the bulk of the cross-maturity variation.

⁵The approximately equal loadings of the debt expansion shock across the four maturities reflect the structure of the principal component decomposition and the normalization imposed by Bi et al. (2026), who restrict this factor to load with equal incidence across the 2-, 5-, 10-, and 30-year segments. This does not imply that yields rise by the same amount at each maturity in response to the underlying issuance surprise; rather, it means that the *identified and normalized* shock units are defined to have symmetric loadings. This is a consequence of data limitations: Phillot (2025) makes available only the normalized shock series, and a more granular maturity-specific identification is not feasible with the published data. Nonetheless, the first principal component explains approximately 80 percent of the cross-maturity variation in within-window yield changes, suggesting that the level shift captured by this factor accounts for the dominant share of the common movement and that the imposed restriction is a reasonable approximation.

Figure 1: U.S. Treasury auction supply shocks



Note: Time series of U.S. Treasury auction supply shocks (debt expansion and maturity extension). Positive values indicate contractionary surprises (upward yield surprises). The original public sample of Phillot (2025) spans 10/1998–01/2020 ($n = 462$); the shaded region marks the post-2020 extension constructed from Bloomberg intraday tick data under the identical identification (Appendix A). Decomposition identifies top two PCA factors and then applies transformation that restricts Debt expansion shock to have same incidence on all four maturities and Maturity extension shock to have sum-zero effect on four maturities.

LSAP component of U.S. monetary policy surprises from Swanson (2021). The standard deviation of the on-impact 10-year yield response to a one-standard-deviation LSAP shock is approximately 4 times larger than that of the debt expansion shock. However, LSAP shocks occur only at scheduled FOMC meetings—eight per year—whereas Treasury auction announcements occur roughly 35–40 times per year. When aggregated to monthly frequency using the Gertler and Karadi (2015) weighting scheme, the cumulative within-month yield variation attributable to Treasury supply shocks is therefore of a comparable order of magnitude to that from LSAP shocks. Moreover, because Treasury auctions are more frequent, the monthly aggregated supply shock series exhibits less lumpiness and greater coverage across calendar months, which improves the precision of monthly local projection estimates. This comparison underscores that, while individual Treasury supply surprises are small, their cumulative effect is economically significant when considered over the relevant aggregation horizon.

2.2 Financial and macroeconomic outcomes

I construct two outcome panels:

Daily sovereign yields. Daily zero-coupon or benchmark 10-year sovereign bond yields are obtained from Haver Analytics for up to 39 economies (an unbalanced panel). For the United States, I use the on-the-run 10-year Treasury yield. All series are expressed in percentage points, aligned to U.S. trading days, and nontrading days are excluded. The absolute majority of sovereign yield series in my panel are denominated in local currency; in the small number of cases where a local-currency series was unavailable or insufficiently liquid over the sample period, I use the

sovereign benchmark available in the Haver database for that country.⁶

Monthly industrial production. Monthly industrial production (IP) indices are obtained from Haver Analytics for up to 58 advanced and emerging market economies (an unbalanced panel). Series are seasonally adjusted, expressed in logarithms, and growth rates computed as log differences. CPI is yoy monthly series from the IMF WEO database. For the United States, the monthly analysis also includes core macro-financial indicators (employment, short-term interest rates, the 10-year Treasury yield, equity prices, and the government debt-to-GDP ratio) to characterize the domestic transmission mechanism. Table 1 documents country coverage and the first year of availability for each variable across the panel.

2.3 Country characteristics for heterogeneity analysis

To study heterogeneous spillovers, I compile annual country-level characteristics that proxy for financial-market exposure, macroeconomic resilience, and market access:

- *Foreign investor share*: the fraction of outstanding general government marketable debt held by nonresidents, from the Arslanalp–Tsuda database (Arslanalp and Tsuda, 2014a,b), covering 63 countries.
- *Capital account openness*: the Chinn–Ito index, capturing legal and de facto cross-border capital mobility.
- *Fiscal space*: a summary measure of the government’s buffer to absorb shocks through fiscal policy.
- *Current account balance*: ratio to GDP, capturing reliance on external financing.
- *Debt-at-risk*: a tail risk measure for the government debt trajectory.
- *U.S. interest rate beta*: the sensitivity of each country’s 10-year sovereign yield to global risk conditions, measured as the 31-day rolling correlation between the country’s 10-year yield and the VIX index. Countries with a higher beta have yields that are more responsive to global risk sentiment.
- *Exchange rate regime*: classified as floating or freely floating based on the IMF’s Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER) de facto classification.

⁶This applies primarily to a small subset of emerging markets where the local-currency sovereign bond market was either nascent or illiquid during the sample. In all such cases, the series used is the sovereign benchmark in HAVER, which may be denominated in a foreign currency. For the vast majority of countries in the panel, the 10-year yield is a local-currency sovereign instrument.

Table 1: Data coverage: first year of availability by country and variable

Country	10-year bond yield	Industrial Production	Capital Openness	Foreign inv. share	Bid/Ask spread	Current Account/GDP	Debt to GDP	Debt to 10y spread	Cyc. Adj. Primary Bal.	Fiscal pro- cyclicality	Fiscal space	Debt at risk	NBFI share	Float (0/1)	VIX sensitivity	US trade share	GDP per capita
ARG	1976	1974	1970	1970	2003	1980	1990	2014	1993	1921	2010	1992	1989	1999	1999	1985	1980
AUS	1994	2005	1970	1995	2014	1980	1988		1991	1921	2006	1990	2003	1999	1990	1985	1980
AUT			1970	1995		1980	1988		1995	1921	2006	1988	1995	1999	1994	1985	1980
AZE	1995	2012	1996	1995		1992	1995			1921	2006	1995	2001	1999	1995	1985	1980
BEL		2000	1970	1995	2014	1980	1986	2014	1980	1921	2006	1986	1995	1999	1995	1985	1980
BGR		2000	1994	1994		1980	1998		2000	1921	2006	1998	1995	1999	1992	1980	1980
BIH		2006	1999	1999		1998	1998		1999	1999	2006	1998	2001	1999	1994	1996	1996
BLR		2008	1996	1994		1992	2004		2010	1921	2004	2001	2001	1999	2006	1992	1990
BRA	2006	2002	1970	1989	2014	1980	2001	2014	2001	1921	2006	2001	2001	1999	2006	1990	1980
CAN	1989	1981	1970	1998	2014	1980	1986	2014	1979	1921	2006	1986	2001	1999	1990	1985	1980
CHE	1994	2004	1996	1999		1980	1990		1983	1921	2006	1990	1999	1999	1994	1985	1980
CHL	2004	2009	1970	2003		1980	2007		2001	1921	2006	2007	2003	1999	2004	1985	1980
CHN	2006	1997	1984	1995	2014	1997	2000	2014	1990	1921	2009	2000	1995	1999	2006	1985	1980
CYP		2000	1970	1995		1980	1995		1995	1921	2006	1995	2008	1999	2000	1992	1980
CZE	2000	2000	1996	1995		1995	1995		1995	1921	2006	1995	1995	1999	2000	1995	1995
DEU	1994	1952	1970	1991	2014	1980	1991	2014	1991	1921	2006	1991	1998	1999	1994	1985	1980
DNK	1995	2021	1970	1995	2014	1980	1992	2014	1986	1921	2006	1992	1997	1999	1995	1985	1980
EGY		2013	1970	2001		1980	1998		2006	1921	2006	1998	2001	1999	1995	1985	1980
ESP	1989	1992	1970	1995	2014	1980	1986	2014	1995	1921	2006	1986	1995	1999	1990	1985	1980
EST		2000	1996	1995		1993	1995		2000	1921	2006	1995	1995	1999	1993	1985	1980
FIN	1994	1995	1970	1995		1980	1986		1979	1921	2006	1986	1995	1999	1994	1985	1980
FRA	1994	1970	1970	1995	2014	1980	1986	2014	1980	1921	2006	1986	1995	1999	1994	1985	1980
GBR	1994	1948	1970	1989	2014	1980	1986	2014	1981	1921	2006	1986	1998	1999	1994	1985	1980
GRC	2018	1995	1970	2003		1980	1986		1988	1921	2006	1988	2003	1999	2018	1985	1980
HKG	2010		1970		2014	1980	2005	2014	2005	1921	2006	2005	2005	1999	2010	1985	1980
HRV		1998	1996	1995		1992	1998		2000	1921	2006	1998	1999	1999	1992	1985	1980
HUN	1999	1993	1986	1989	2014	1980	1999	2014	2000	1921	2006	1999	1995	1999	1999	1985	1980
IDN	2010	2010	1970	1998	2014	1980	2003	2014	2004	1921	2006	2003	2001	1999	2010	1980	1980
IND	1998	1971	1970	1997	2014	1980	1991	2014	1996	1921	2006	1991	1997	1999	1998	1985	1980
IRL	1998	1980	1970	2002	2014	1980	1995	2014	1997	1921	2006	1995	2002	1999	1998	1985	1980
ISR	2002		1970	1998		1980	2002		2000	1921	2006	2002	2001	1999	2002	1980	1980
ITA	1995	1980	1970	1989	2014	1980	1986	2014	1988	1921	2006	1986	1989	1999	1995	1985	1980
JOR		2002	1970	1989		1980	1986		1999	1921	2006	1988	2014	1999	1992	1980	1980
JPN	1986	2008	1970	2003	2014	1980	1986	2014	1994	1921	2006	1986	2003	1999	1990	1985	1980
KAZ		1998	1996	1993		1992	2000		2002	1921	2006	2002	2001	1999	1992	1985	1980
KOR	2013	1980	1970	1994	2014	1980	2002	2014	1995	1921	2010	2000	1994	1999	2013	1980	1980
LKA		2010	1970	1989		1980	1990		2001	1921	2010	1999	2001	1999	1999	1985	1980
LIT		1998	1996	1995		1995	1998		2005	1921	2010	1998	1995	1999	1995	1995	1995
LUX		1970		1995		1995	1995		1995	1921	2006	1995	1995	1999	1992	1980	1980
LVA		2000	1996	1995	2014	1992	1998		2002	1921	2006	1998	2004	1999	1992	1980	1980
MEX	2007		1970	1998		1980	2001	2014	1997	1921	2006	2001	2001	1999	2008	1980	1980
MNE		2001		2005		2001	2002					2002	2013	1999	2000	2007	2000
MYS	2019	2015	1970	1989	2014	1980	1990	2014	1991	1921	2006	1990	2001	1999	2019	1985	1980
NLD	1990	2000	1970	1995	2014	1980	1995	2014	1995	1921	2006	1995	1995	1999	1990	1985	1980
NOR	1994	1960	1970	2003	2014	1980	1986	2014	1980	1921	2006	1986	2003	1999	1994	1985	1980
NZL		1987	1970	1993		1980	1986		2005	1921	2006	1986	2001	1999	1999	1985	1980
PER	2005		1970	1999		1980	2005		2000	1921	2006	2005	2004	1999	2005	1980	1980
PHL	2011		1970	1998	2014	1980	1996	2014	1990	1921	2010	1996	2001	1999	2011	1980	1980
POL	1999	1999	1986	1990	2014	1980	1995	2014	1995	1921	2006	1995	1999	1999	1999	1985	1980
PRT	1994	2005	1970	1995	2014	1980	1990	2014	1995	1921	2006	1990	1998	1999	1994	1985	1980
QAT		2016	1976	1990		1980	1990		1991	1921	2006	1990	1990	1999	1999	1992	1980
ROU		2000	1976	1996		1980	2001		2004	1921	2006	2001	1996	1999	1992	1980	1980
RUS		2005	1996	1992		1992	1998		1999	1921	2006	1998	2000	1999	1992	1980	1980
SGP	1998		1970		2014	1980		2014		1921	2010	1991	2001	1999	1998	1980	1980
SLV		2005	1970	1991		1980	1991		2000	1921	2010	1991	2001	1999	1985	1980	1980
SRB		1994		2000		1997	2000		2005		2006	2000	2001	1999	2007	1999	1999
SVK		1998	1996	1995		1993	1995		1995	1921	2006	1995	2008	1999	1993	1993	1993
SVN		2008	1996	1995		1992	1995		1996			1995	1995	1999	1992	1992	1992
SWE		2000	1970	1995	2014	1980	1993	2014	1993	1921	2006	1993	1997	1999	1992	1985	1980
SYC	1992	2013	1981	1989		1980	1990		2000	1921	2006	1990	2001	1999	1992	1985	1980
THA	2001		1970	1998	2014	1980	2000	2014	2000	1921	2010	2000	2001	1999	2001	1980	1980
TUR	2012	1986	1970	1989	2014	1980	2000	2014	2000	1921	2007	2000	2001	1999	2012	1980	1980
TWN		1996				1984											
UKR		2006	1996	1995		1992	1997		2003	1921	2006	1997	2001	1999	1992	1980	1980
USA	1962	1921	1970		2014	1980	1942		2001	1921	2006	1986		1999	1990	1992	1991
ZAF	2019	1993	1970	1998	2014	1980	2000	2014	2000	1921	2010	2000	2001	1999	2019	1985	1980

Note: First calendar year each variable is available for each country in the estimation panel; blank = unavailable. Only countries entering at least one regression (a 10-year yield or industrial production series) are shown. Heterogeneity variables are those used in the interaction regressions.

3 Methodology

3.1 Daily local projections: U.S. and panel yields

I estimate daily local projections (LPs) (Jordà, 2005) for each horizon $h = 0, 1, \dots, 10$ business days. Throughout, let $\Delta_h Z_t \equiv Z_{t+h} - Z_{t-1}$ denote the cumulative h -period change in a variable Z from the day before the shock to h days after. The dependent variable in all daily regressions is therefore the cumulative change in the 10-year sovereign yield from day $t - 1$ to day $t + h$.

U.S. specification. For each horizon h , I estimate:

$$\Delta_h Y_t^{\text{US}} \equiv Y_{t+h}^{\text{US}} - Y_{t-1}^{\text{US}} = \beta_h s_t + \delta' D_t + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + \varepsilon_{t,h}, \quad (1)$$

where s_t is the Treasury supply shock (debt expansion or maturity extension), D_t collects auction and announcement controls together with contemporaneous and lagged values of control variables selected by Granger causality pre-testing (including two lags each of the U.S. 10-year yield, the U.S. short rate, the shock itself, and the Japanese 10-year yield), and $\lambda_{\text{Year}(t)}$ and $\mu_{\text{DOW}(t)}$ are year and day-of-week fixed effects. Standard errors are HAC with Newey–West bandwidth of $h + 1$. The estimated sequence $\{\hat{\beta}_h\}_{h=0}^{10}$ traces the impulse response of the U.S. 10-year yield.

Panel specification. For country i and horizon h :

$$\Delta_h Y_{i,t} \equiv Y_{i,t+h} - Y_{i,t-1} = \beta_h s_t + \phi \Delta_1 Y_{i,t-1} + \delta' D_t + \alpha_i + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + \varepsilon_{i,t,h}, \quad (2)$$

with country fixed effects α_i absorbing time-invariant country heterogeneity. Standard errors are two-way clustered by country and by time (shock date).

3.2 IV normalization: spillover elasticity per unit U.S. yield movement

To express foreign yield spillovers in a unit-free way, as the response of a foreign 10-year yield per basis point increase in the U.S. 10-year yield induced by the shock, I implement a two-stage IV procedure.

First stage. I estimate the on-impact ($h = 0$) U.S. yield response to the supply shock:

$$\Delta_0 Y_t^{\text{US}} \equiv Y_t^{\text{US}} - Y_{t-1}^{\text{US}} = \pi s_t + \delta' D_t + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + v_t. \quad (3)$$

The fitted value $\widehat{\Delta_0 Y_t^{\text{US}}} = \hat{\pi} s_t$ isolates the component of U.S. yield variation attributable to the supply shock. The first-stage HAC F-statistic is 26.0, confirming instrument strength.

Second stage. For each horizon h , I substitute $\widehat{\Delta_0 Y_t^{\text{US}}}$ for s_t in the panel LP:

$$\Delta_h Y_{i,t} = \theta_h^{\text{IV}} \widehat{\Delta_0 Y_t^{\text{US}}} + \delta'_h D_t + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + \varepsilon_{i,t,h}. \quad (4)$$

The coefficient θ_h^{IV} is the *spillover elasticity*: the change in a foreign 10-year yield at horizon h associated with a 1 basis point shock-induced increase in the U.S. 10-year yield on the announcement day.

A concern is that Treasury auctions are “regular and predictable” and therefore largely *anticipated*: the narrow within-window move captures only the unexpected sliver of supply news, not the full effect of issuing a given quantity of debt. A reader may then ask what the effect of a “pure” auction would be—say, an unanticipated decision to issue an extra billion dollars. This question, while legitimate, concerns a different estimand than mine and does not threaten my results. The structural *level* effect of issuance quantity on U.S. yields is precisely the object that Phillot (2025) and Bi et al. (2026) identify—the former instrumenting with the auction offer amount—and I take that domestic literature as given. My estimand is instead the *spillover elasticity* θ_h : the response of foreign yields per basis point of shock-induced movement in the U.S. yield, with the supply shock used purely as an *instrument* for the U.S. repricing. The magnitude of the shock is never interpreted as a quantity of issuance. Formally, suppose the within-window surprise captures only a fraction κ of the “true” supply news (because of anticipation, the unit normalization, or tick-level measurement error). Because the first-stage loading and the reduced-form foreign response are both coefficients on the *same* instrument s_t , κ scales the two identically and cancels in the IV ratio $\theta_h = \hat{\rho}_h / \hat{\pi}$, which is therefore invariant to κ and to classical measurement error in s_t . Anticipation shrinks the *size* of the instrument—a relevance question, answered by the first-stage F -statistic of 26—but not its *validity*, which requires only that s_t be exogenous (Section 2.1) and affect foreign yields solely through the global repricing of U.S. long rates. Consistent with this logic, replacing the supply shock with the demand-side LSAP shock of Swanson (2021)—a different identification, with a different anticipation structure—yields spillover profiles of similar shape (Section 4.8), exactly as expected if the elasticity reflects bond-market transmission rather than any feature of how the instrument is generated.

3.3 Decomposing the U.S. yield channel: risk-free rate versus term premium

A natural question is whether the international transmission of U.S. supply shocks operates through the *expected risk-free rate* (RA) component of the U.S. 10-year yield or through its *term-premium* (TP) component—that is, which segment of the U.S. curve foreign markets actually arbitrage. Because I observe the Adrian et al. (2013)-style decomposition of the U.S. 10-year yield into these two pieces, I can answer this directly by replacing the single first stage of Section 3.2 with two parallel just-identified first stages that split the on-impact U.S. move:

$$\Delta_0 \text{RA}_t^{\text{US}} = \pi^{\text{RA}} s_t + \delta' D_t + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + v_t^{\text{RA}}, \quad \Delta_0 \text{TP}_t^{\text{US}} = \pi^{\text{TP}} s_t + \delta' D_t + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + v_t^{\text{TP}}. \quad (5)$$

Since the decomposition is additive, $\text{RA}_t^{\text{US}} + \text{TP}_t^{\text{US}} = Y_t^{\text{US}}$, the first-stage loadings satisfy $\pi^{\text{RA}} + \pi^{\text{TP}} = \pi$, so the ratio π^{TP} / π measures the share of the shock-induced U.S. repricing that is term-

premium-driven. For each component $k \in \{\text{RA}, \text{TP}\}$ I then estimate the second-stage panel LP

$$\Delta_h Y_{i,t} = \theta_h^k \widehat{\Delta_0 k_t^{\text{US}}} + \delta'_h D_t + \lambda_{\text{Year}(t)} + \mu_{\text{DOW}(t)} + \varepsilon_{i,t,h}, \quad (6)$$

where $\widehat{\Delta_0 k_t^{\text{US}}} = \hat{\pi}^k s_t$. The coefficient θ_h^k is the foreign 10-year yield response per basis point of shock-induced movement in the U.S. component k , exactly mirroring the financial-channel logic of Section 4.7 but isolating which leg of the U.S. curve foreign markets import. Standard errors are two-way clustered by country and time.

3.4 Monthly local projections: macro effects and output spillovers

I aggregate daily shocks to monthly frequency by summing over a rolling 31-day window following Gertler and Karadi (2015), where each daily observation receives a weight proportional to the number of days remaining in the window, reflecting the intuition that older shocks have had more time to influence monthly outcomes.⁷ I then estimate monthly LPs over horizons $h = 0, 1, \dots, 24$ months. Throughout, let $\Delta_h y_{i,t} \equiv y_{i,t+h} - y_{i,t-1}$ denote the cumulative h -month change in outcome y for country i from the month before the shock. For monthly outcome $y_{i,t}$ (log IP, log CPI, etc.):

$$\Delta_h y_{i,t} \equiv y_{i,t+h} - y_{i,t-1} = \beta_h s_t + \sum_{k=1}^2 \Gamma_k X_{i,t-k} + \sum_{k=1}^2 \delta_k s_{t-k} + \alpha_i + \lambda_t + \varepsilon_{i,t,h}, \quad (7)$$

where $X_{i,t-k}$ collects lagged macro-financial controls, including lagged log IP, log CPI, interest rates, and Granger-causality-selected additional controls, and s_{t-k} are two lags of the shock. For the U.S. domestic regression, country fixed effects α_i are omitted; for the panel, both country fixed effects α_i and time fixed effects λ_t are included as specified. Standard errors are Newey–West HAC with a bandwidth of $h+1$ for the U.S. single-country regressions, and two-way clustered by country and time for the panel specifications.

3.5 Channel decomposition

To characterize the relative importance of the financial, trade, and exchange rate channels, I implement three complementary, exposure-based identification designs and report the resulting structural impulse response functions. I stress that these designs are *not* a strict econometric variance decomposition: the three mediators are correlated across recipient countries and are not mutually orthogonal, so they cannot be read as an exact apportionment of the total spillover. I therefore interpret them as complementary and suggestive evidence on the *conduits* of transmission, and I note that each design is built on a cross-country exposure interaction of exactly the kind I exploit systematically in the heterogeneity analysis of Section 3.6.

⁷Results are robust to an alternative aggregation scheme in which daily shocks are simply summed within each calendar month without reweighting by the number of remaining days. The impulse response functions under this simpler aggregation are qualitatively and quantitatively similar to my baseline, with point estimates falling within the confidence bands of the main specification at all horizons.

Financial channel. The financial channel operates through shock-induced increases in U.S. long-term yields tightening global financial conditions and raising borrowing costs for foreign firms and sovereigns. For each country i , I estimate a daily local projection over horizons $h = 0, \dots, 10$ days and select the country-specific peak horizon $h_i^* = \arg \max_h |t_i^h|$, where t_i^h is the t -statistic of the shock coefficient at horizon h . This *most powerful* selection criterion accounts for the fact that estimation precision varies across horizons. The country-specific daily first-stage coefficient $\hat{\beta}_i^{h^*}$ is then used to construct a country-varying shock series $\hat{\beta}_i^{h^*} \cdot s_t$ on event days, which is aggregated to monthly frequency via the Gertler–Karadi weighting scheme. The resulting panel of GK-aggregated financial shocks serves as the mediator in a second-stage panel LP for foreign IP, with country fixed effects and two-way clustered standard errors. This design identifies the component of foreign output spillovers attributable to the global repricing of long-term rates.

Trade channel. The trade channel operates through the reduction in U.S. aggregate demand, which lowers foreign export revenues. In order to separately identify the country-specific exposure to the raw IP contraction in the U.S., I exploit cross-country heterogeneity in bilateral trade exposure. Specifically, the mediator for country i at time t is

$$m_{i,t}^{\text{trade}} = \text{USTS}_{i,t} \times s_t,$$

where $\text{USTS}_{i,t}$ denotes the bilateral U.S. trade share of country i 's GDP (exports to the United States, as a percent of GDP, sourced from U.S. Census trade data and WEO nominal GDP). This mediator is genuinely country-varying: economies more exposed to U.S. demand receive a larger effective shock, providing cross-sectional identification distinct from the pooled shock. The second-stage coefficient $\hat{\beta}^{\text{trade}}$ measures the response of foreign IP per unit of trade-exposure-weighted shock, and the contribution at average exposure is recovered as $\bar{\gamma}^{\text{trade}} \times \hat{\beta}^{\text{trade}}$, where $\bar{\gamma}^{\text{trade}} = \overline{\text{USTS}}$ is the cross-country mean trade share.

Exchange rate channel. The exchange rate channel operates through the appreciation of the U.S. dollar following a bond supply shock, which acts on foreign output through two channels of opposing sign. A competitiveness channel works in the expansionary direction: a weaker domestic currency raises net exports and supports foreign activity. A balance-sheet channel works in the contractionary direction: currency depreciation tightens financial conditions for dollar-indebted firms and sovereigns, compressing investment. Because these forces offset, the net effect on foreign output is ambiguous a priori, and a weak or insignificant net response is fully consistent with both channels being individually operative. I identify this channel using the same daily peak-horizon approach as the financial channel, but with outcome variable $100 \cdot \log(\text{FX}_{i,t})$, so that the estimated coefficient $\hat{\beta}_i^{h^*}$ is in units of percent bilateral exchange rate change per shock unit. Country-varying daily first-stage series are GK-aggregated to monthly frequency and enter a second-stage panel LP for foreign IP. The resulting coefficient captures the IP response per one-percent shock-induced currency movement.

3.6 Heterogeneity: interacted local projections with time fixed effects

To assess the role of country characteristics in amplifying or dampening spillovers, I estimate interacted LPs with time fixed effects:

$$\Delta_h y_{i,t} = \beta_h s_t \cdot \tau_{i,t} + \gamma_h \Delta_1 y_{i,t-1} + \delta'_h D_{i,t} + \alpha_i + \lambda_t + \varepsilon_{i,t,h}, \quad (8)$$

where s_t is the U.S. supply shock (or the IV-instrumented U.S. yield change $\widehat{\Delta_0 Y_t^{\text{US}}}$), $\tau_{i,t}$ is the country characteristic of interest (e.g., foreign investor share, capital account openness, fiscal space), α_i are country fixed effects, and λ_t are time fixed effects. Because time fixed effects absorb the common component of s_t , identification of β_h comes entirely from cross-sectional variation in $\tau_{i,t}$ within the same time period. I scale $\hat{\beta}_h$ by the interquartile range (IQR) of $\tau_{i,t}$ across countries, so that the reported coefficient represents the *differential spillover* for a country at the 75th percentile of the characteristic relative to a country at the 25th percentile.

4 Results

4.1 Daily yield responses: domestic impact

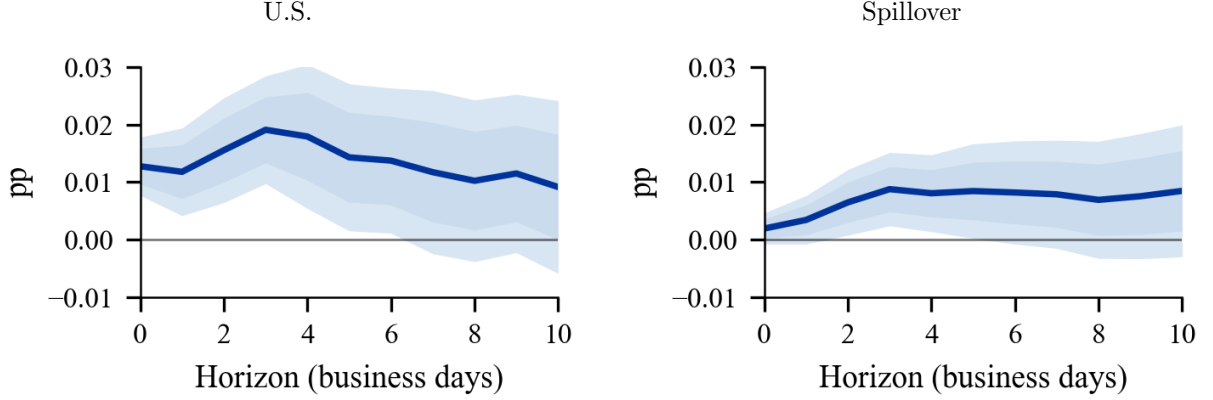
A one standard deviation debt expansion shock raises the U.S. 10-year Treasury yield by approximately 1.2 basis points on impact (Figure 2, left panel). The response builds over the first four trading days and persists through the full 10-day horizon. Results suggest that Treasury auction announcements convey durable information about future supply conditions that markets continue to price in over several days.

4.2 Daily yield responses: international spillovers

The same shock transmits strongly to foreign sovereign bond markets (Figure 2, right panel). Foreign 10-year yields rise by approximately 1 basis point within three days of the shock, with the response stabilizing thereafter. Normalizing by the shock-induced U.S. yield response using the IV approach of Section 3.2, the implied spillover elasticity reaches approximately 0.7 by day three (Figure 3). In other words, a 10-basis-point increase in the U.S. 10-year yield causes the foreign panel yield to rise by approximately 7 basis points within three business days.

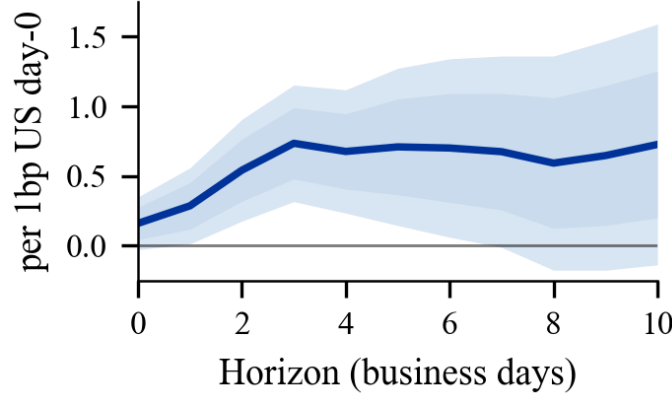
These results indicate a high degree of integration between U.S. and global sovereign bond markets. The elasticity, on the order of 0.7, is substantial: it implies that foreign government borrowing costs absorb a large share of a U.S. supply shock, consistent with global capital markets treating U.S. Treasuries and other high-grade sovereign bonds as close substitutes whose yields are continuously arbitrated.

Figure 2: U.S. and foreign panel 10-year yields response to debt expansion shock



Note: Daily impulse responses of U.S. (left) and foreign panel (right) 10-year sovereign yields in basis points per one-standard-deviation debt expansion shock, estimated by local projections. Shaded bands denote 68% and 90% confidence intervals. Standard errors: HAC (Newey–West, bandwidth $h + 1$) for the U.S. single-country regression; two-way clustered by country and time for the panel. Daily OLS local projections are estimated over horizons of up to ten business days. The dependent variable is $Y_{t+h} - Y_{t-1}$ for the U.S. 10-year Treasury yield and $Y_{i,t+h} - Y_{i,t-1}$ for foreign 10-year sovereign yields. The shock is a contractionary Treasury-supply shock identified on auction announcement days as the first principal component raising the yield curve and residualized ex-ante with respect to fixed effects (year, day-of-week, and country, where applicable), auction related dummies, and controls selected based on Granger-causality tests.

Figure 3: 10-year yields spillover elasticity

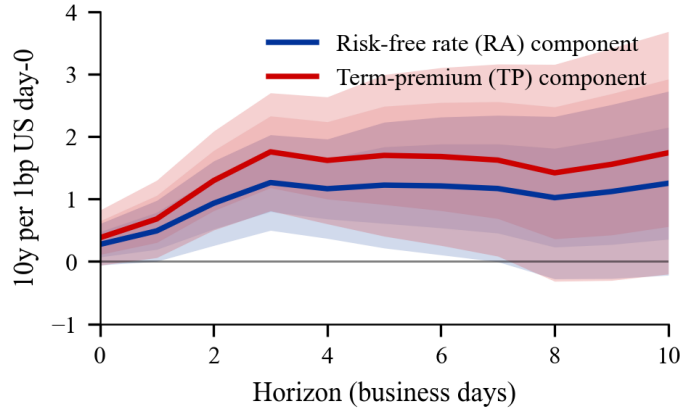


Note: Global spillover to foreign 10-year sovereign yields per 1 basis point shock-induced increase in the U.S. 10-year yield (IV normalization). First-stage HAC F-statistic = 26.0. Shaded bands denote 68% and 90% confidence intervals. Standard errors: two-way clustered by country and time. Daily OLS local projections are estimated over horizons of up to ten business days. The first stage dependent variable is $Y_t - Y_{t-1}$ for the U.S. 10-year Treasury yield and second stage is $Y_{i,t+h} - Y_{i,t-1}$ for foreign 10-year sovereign yields. The shock is a contractionary Treasury-supply shock identified on auction announcement days as the first principal component raising the yield curve and residualized ex-ante with respect to fixed effects (year, day-of-week, and country, where applicable), auction related dummies, and controls selected based on Granger-causality tests.

4.3 Which U.S. yield component transmits: risk-free rate versus term premium

Applying the two-first-stage IV of Section 3.3, I find that the international spillover is driven predominantly by the *term-premium* component of the U.S. 10-year yield rather than the expected risk-free rate component (Figure 4). The term premium accounts for the larger share of the shock-induced U.S. move, consistent with the preferred-habitat interpretation in Bi et al. (2026), in which a supply surprise raises the compensation arbitrageurs require for bearing duration risk. The second-stage elasticity of foreign yields with respect to a shock-induced basis point of the U.S. term premium is large and builds over the first few days, whereas the elasticity with respect to the risk-free rate component is smaller and less precisely estimated. The reading is intuitive: the supply shock is, by construction, a duration-risk-repricing event, and it is precisely this term-premium repricing of U.S. Treasuries that global capital markets arbitrage into foreign long rates. This complements the financial-channel evidence of Section 4.7 by identifying the specific segment of the U.S. yield—the term premium—through which the financial channel operates.

Figure 4: Spillover by U.S. 10-year component: risk-free rate vs. term premium



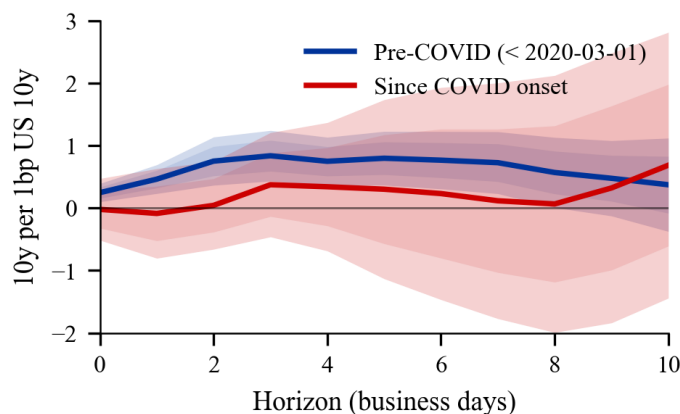
Note: Foreign 10-year sovereign yield response per 1 basis point of shock-induced movement in the risk-free rate (RA) and term-premium (TP) components of the U.S. 10-year yield, from the two just-identified first stages of Eq. (5) and the second stage of Eq. (6). The U.S. 10-year decomposition follows Adrian et al. (2013). Shaded bands denote 90% confidence intervals. Standard errors: two-way clustered by country and time. Daily local projections over horizons of up to ten business days.

4.4 Spillovers since the onset of COVID

The extended shock series (Section 2.1) lets us ask whether the strength of the financial spillover has changed in the post-COVID environment of elevated debt and global fiscal uncertainty. I re-estimate the daily IV spillover elasticity of Section 3.2 separately on the pre-COVID sample and on the sample since the onset of COVID (Figure 5). I restrict this comparison to the financial (yield) spillover and deliberately do *not* repeat the monthly IP or CPI local projections on the post-COVID subsample: the macro panel since the COVID onset is too short to estimate 12–24-month horizon responses with any reliability. The post-COVID subsample is short, so the elasticity is

estimated with much wider confidence bands than the pre-COVID path. The point estimates since the COVID onset are positive at most horizons but somewhat lower than the pre-COVID estimates, and the wide post-COVID confidence bands comfortably encompass the pre-COVID path at every horizon. I therefore cannot reject that the strength of transmission has been unchanged, and I caution against reading the lower point estimates as evidence of weakening given the imprecision. The first-stage instrument remains strong in both subsamples. The reasonable conclusion is that U.S. supply shocks continued to pass through to foreign long rates in the higher-issuance post-pandemic period, with no statistically discernible change in magnitude relative to the pre-COVID sample.

Figure 5: Daily yield spillover elasticity: pre-COVID vs. COVID onset



Note: IV-normalized spillover to foreign 10-year sovereign yields per 1 basis point shock-induced increase in the U.S. 10-year yield, estimated separately on the pre-COVID sample and the sample since the onset of COVID. The comparison is restricted to the financial (yield) spillover; monthly IP/CPI responses are not re-estimated on the post-COVID subsample because it is too short for medium-horizon local projections. The post-COVID elasticity is imprecisely estimated; its confidence band encompasses the pre-COVID path at all horizons. Shaded bands denote 90% confidence intervals. Standard errors: two-way clustered by country and time.

4.5 Monthly U.S. macro responses

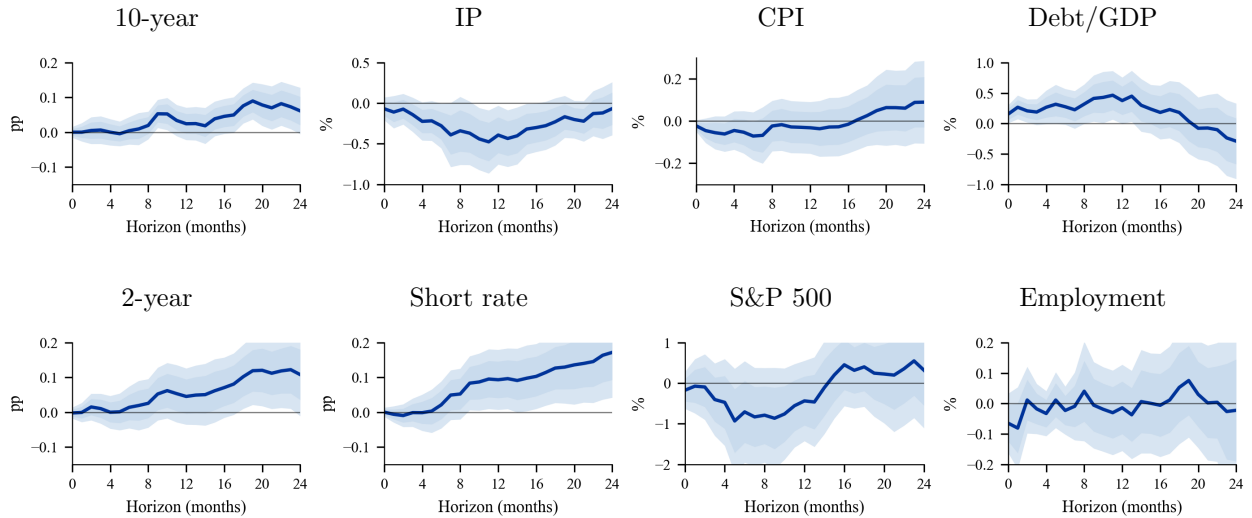
At monthly frequency, the aggregated debt expansion shock generates a tightening in U.S. financial conditions consistent with its yield-raising effect: the 10-year Treasury yield rises, and the short rate and 2-year yield rise. On the real side, U.S. industrial production declines persistently, reaching a trough of approximately -0.4 percent at an eight-month horizon before reverting toward baseline (Figure 6). Inflation, as measured by the CPI, shows a mild decline, consistent with the contractionary demand effect. Employment falls gradually, peaking at 9-10 months. The government debt-to-GDP ratio rises in the first 12 months.

An important interpretive point is that the monthly impulse response functions capture the *general equilibrium* response of the economy to the identified supply shock. This means that any endogenous monetary policy reaction—for instance, the Federal Reserve adjusting the short rate

in response to the tightening in financial conditions or the deterioration in real activity—is part of the transmission mechanism and is reflected in the estimated IRFs by construction. Crucially, the Treasury supply shocks are identified at high frequency from narrow event windows around auction announcements and are by construction unpredictable (as confirmed by the Granger causality tests reported in Section 2.1). Moreover, all specifications include the contemporaneous LSAP and federal funds rate components of Swanson (2021)’s monetary policy surprises as controls, ensuring that the estimated effects of Treasury supply shocks are not contaminated by coincident monetary policy actions. The IRFs therefore trace the full macroeconomic propagation of an exogenous supply surprise, inclusive of all endogenous responses, including those of the central bank.

These domestic results are closely aligned with the benchmark estimates of Bi et al. (2026) (Figure 7), confirming the validity of my specification and aggregation approach. The strong correspondence across the 10-year yield, CPI, and industrial production responses in particular provides an important external validation.

Figure 6: U.S. domestic response to debt expansion shock



Note: U.S. monthly impulse responses to an aggregated debt expansion shock. Responses are cumulative and expressed in percentage points (for interest rates) or percent (for log variables) to the one-standard-deviation first principal component raising the yield curve after Treasury auction announcement shock, aggregated at the monthly frequency. Shaded areas denote 68% and 90% confidence bands. Estimates are obtained using OLS local projections at monthly frequency, for horizons up to 24 months. Standard errors: HAC (Newey–West, bandwidth $h + 1$). Regressions control for lagged dependent variables, lagged shocks, key macroeconomic conditions (including interest rates, activity, and prices), and Granger-causality informed variables.

Plausibility of the magnitudes. A natural question is whether the magnitudes are plausible: a one standard deviation shock raises 10-year yields by only 1–2 basis points on impact, yet is associated with an industrial production contraction of around 0.4 percent at peak. This ratio of large real effects from a small yield movement is a feature shared with high-frequency identified

monetary policy shocks. Gertler and Karadi (2015), for instance, find that a 25-basis-point surprise tightening in the federal funds rate generates an IP decline of 0.5–1 percent at a 12–18 month horizon, implying a slightly larger elasticity to the short-rates change. Swanson (2021)’s LSAP component, which I revisit in Section 4.8, produces 10-year yield movements and IP responses of broadly comparable relative magnitude to those I document here. Jarociński and Karadi (2020) similarly find that monetary policy surprises of modest high-frequency magnitude—on the order of a few basis points in short-term rate futures—generate economically significant and persistent effects on real activity and prices when the information component is properly separated from the policy component. Taken together, the magnitude of the domestic U.S. response is well within the range documented for monetary policy shocks of similar size, and the alignment with Bi et al. (2026) further corroborates the estimates.

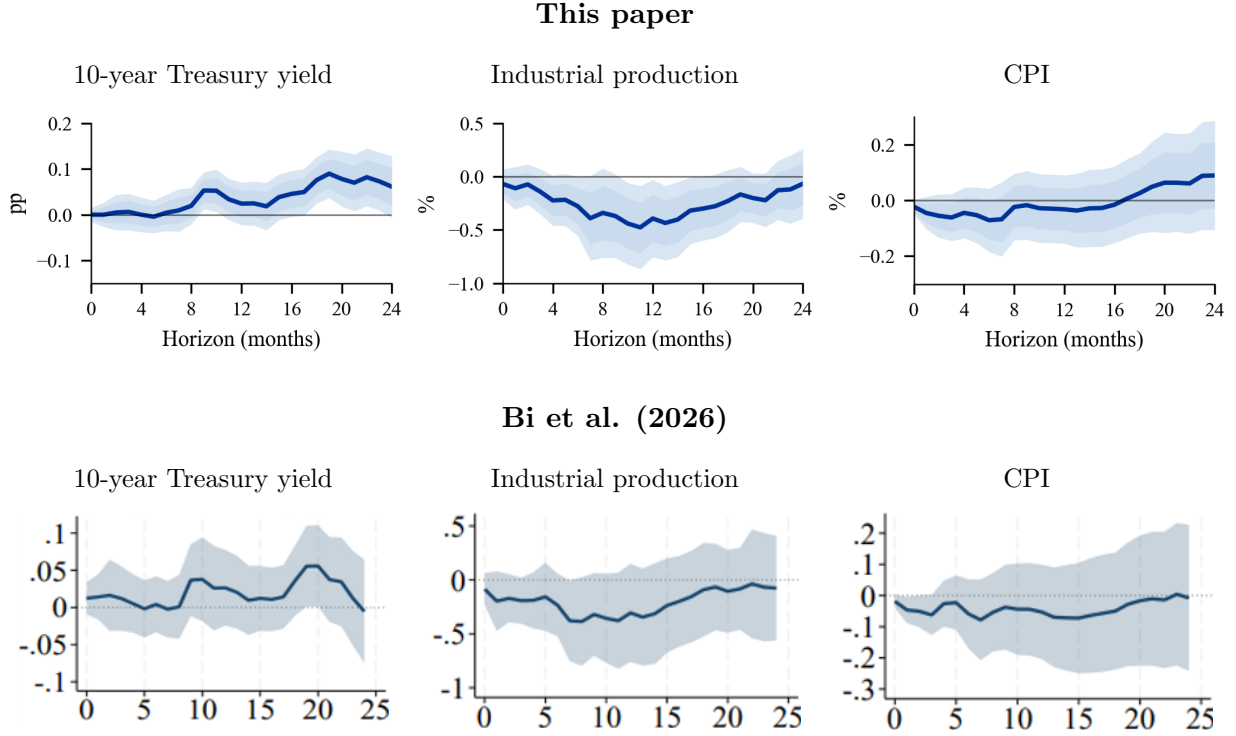
CPI response. The decline in the CPI following the debt expansion shock is consistent with the contractionary mechanism: the shock raises long-term borrowing costs, which compresses aggregate demand and puts downward pressure on prices. The estimated CPI decline of approximately 0.1 percent over the first 12 months is modest but statistically significant, and its timing—lagging the IP decline—is consistent with standard New Keynesian Phillips curve dynamics in which prices adjust with a delay relative to real activity.

4.6 Monthly output spillovers to foreign economies

Monthly panel LPs reveal that the contractionary effects of U.S. bond supply shocks extend well beyond the United States (Figure 8). Foreign industrial production declines by approximately 0.4 percent at a 6–12 month horizon, before partially recovering, with the peak effect statistically significant at the 90 percent level across most specifications. The international spillover is of comparable magnitude to the domestic U.S. impact over the same horizon, but more persistent.

Foreign contraction is as large as the domestic one, rather than a fraction of it, for three reasons. First, the domestic U.S. response already embeds the endogenous reaction of monetary policy discussed in Section 4.5, which leans against the tightening at home; foreign economies, by contrast, import the global repricing of long rates without a domestic policy offset calibrated to a shock they did not originate, so their response is comparatively un-cushioned. Second, the foreign panel response is a composite that aggregates several reinforcing channels—the direct tightening of global financial conditions via higher long-term rates, reduced U.S. import demand, and exchange-rate adjustment—which together sustain a contraction comparable to the domestic impulse despite the absence of a direct issuance effect abroad. Third, the recipient panel is weighted toward financially open and U.S.-trade-exposed economies (Section 4.9), precisely the countries in which these channels compound. I therefore read the foreign response as the expected consequence of an un-offset, multi-channel transmission landing on the most exposed economies.

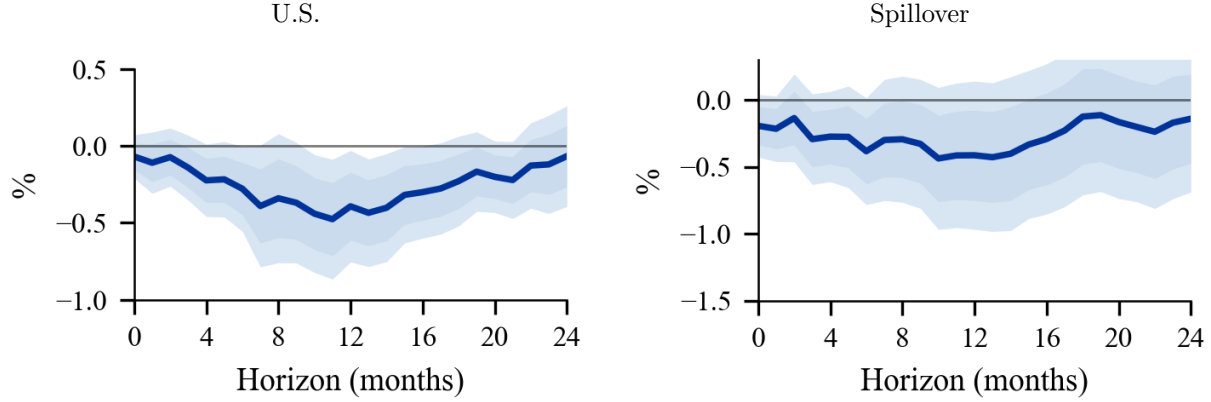
Figure 7: Comparison of U.S. response with benchmark results from Bi et al. (2026)



Note: Comparison of U.S. impulse responses (10-year yield, CPI, industrial production) with benchmark results from Bi et al. (2026). Responses are cumulative and expressed in percentage points (for interest rates) or percent (for log variables) to the one-standard-deviation first principal component raising the yield curve after Treasury auction announcement shock, aggregated at the monthly frequency. Shaded areas denote 68% and 90% confidence bands. Estimates are obtained using OLS local projections at monthly frequency, for horizons up to 24 months. Standard errors: HAC (Newey–West, bandwidth $h + 1$). Regressions control for lagged dependent variables, lagged shocks, key macroeconomic conditions (including interest rates, activity, and prices), and Granger-causality informed variables.

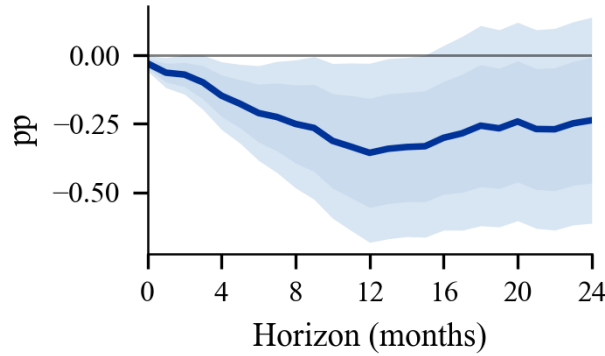
Foreign CPI response. The contractionary effects of U.S. bond supply shocks extend to foreign prices as well. Panel estimates show that the CPI in foreign economies declines by a statistically and economically significant margin over the 6–18 month horizon following a debt expansion shock (Figure 9). The magnitude of the foreign CPI decline—on the order of 0.2–0.3 percent at peak—is consistent with the contraction in foreign industrial production documented above and supports the interpretation that the transmission operates through a demand channel: higher U.S. and global interest rates reduce aggregate demand in foreign economies, which in turn puts downward pressure on consumer prices. The international CPI response thus reinforces the conclusion that U.S. bond supply shocks generate a genuine contractionary impulse abroad, affecting not only real activity but also the price level, consistent with the standard transmission of a financial tightening through lower demand.

Figure 8: U.S. and foreign panel industrial production response to debt expansion shock



Note: Monthly impulse responses of U.S. (left) and foreign panel (right) industrial production to an aggregated debt expansion shock. Shaded bands denote 68% and 90% confidence intervals. The shock is constructed by aggregating high-frequency event-day Treasury supply surprises within each month and set to zero in non-event months. Panel estimates exclude the United States and include country fixed effects and controls. Standard errors: HAC (Newey–West, bandwidth $h + 1$) for U.S.; two-way clustered by country and time for the panel.

Figure 9: Foreign panel CPI response to debt expansion shock



Note: Monthly impulse response of foreign panel CPI to an aggregated debt expansion shock. Shaded bands denote 68% and 90% confidence intervals. Standard errors: two-way clustered by country and time. Panel estimates exclude the United States and include country and time fixed effects.

4.7 Channel decomposition: financial, trade, and exchange rate linkages

Figure 10 reports the structural β impulse response paths for the three exposure-based channel designs. I read these as suggestive evidence on the conduits of transmission rather than as a formal decomposition.

Financial channel. The financial channel is quantitatively important and operates at high frequency. The modal peak horizon across countries is $h^* = 3$ days, with a mean daily first-stage coefficient of $\bar{\gamma}^{\text{fin}} \approx +0.011$ pp of foreign yield per shock unit, confirming near-immediate pass-through of U.S. supply shocks to global long-term rates. The second-stage panel LP shows a significant and persistent decline in foreign IP in response to the instrumented increase in foreign yields, with the response deepening over a 6–10 month horizon before gradually recovering (Figure 10, left panel). The implied elasticity of foreign output with respect to a shock-induced repricing of long-term rates is economically meaningful and of the expected sign, consistent with a borrowing-cost channel operating through both firms’ cost of capital and sovereign fiscal space.

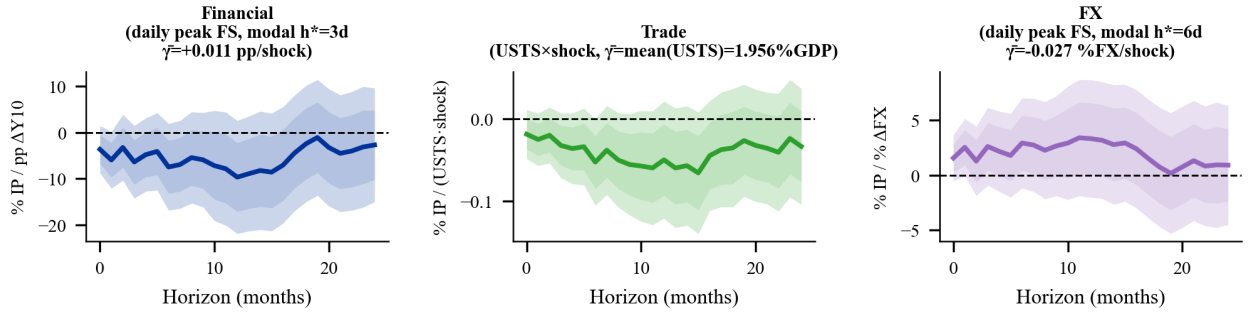
Trade channel. The trade channel provides an independent and complementary source of contractionary spillovers, identified through cross-country variation in bilateral U.S. trade exposure. The mean trade share in the sample is $\overline{\text{USTS}} \approx 1.96\%$ of GDP, with substantial dispersion across countries and over time. The second-stage estimates show a negative and persistent response of foreign IP to the trade-exposure-weighted shock, with the trough deepening through the medium term (Figure 10, center panel). The structural coefficient measures the IP response per unit of ($\text{USTS} \times \text{shock}$); at average trade exposure, the implied contribution is $\bar{\gamma}^{\text{trade}} \times \hat{\beta}^{\text{trade}}$. Because the trade mediator varies across countries, it is identified from cross-country variation in U.S. trade exposure rather than from the common shock, directly connecting the trade conduit to the exposure-based heterogeneity results below.

Exchange rate channel. The exchange rate channel is identified through shock-induced currency movements, with a modal peak horizon of $h^* = 6$ days. In the extended sample, however, the first-stage currency response is small and no longer cleanly signed, and the second-stage IP response with respect to shock-induced FX movements (Figure 10, right panel) is imprecisely estimated and not robustly of the competitiveness sign. I read this as consistent with two offsetting forces operating through the exchange rate—a competitiveness channel that would compress foreign output and a balance-sheet channel for dollar-indebted borrowers—whose net effect this design cannot pin down. I therefore treat the exchange rate as a plausible but, in my data, weakly identified conduit, and rely on the exchange-rate-regime heterogeneity in Section 4.9 for sharper evidence on its role.

Taking stock. The financial and trade conduits both deliver foreign-IP responses of the expected

contractionary sign and of broadly comparable magnitude at medium horizons, while the exchange-rate conduit is weakly identified in my sample. Because each design is an exposure interaction, the recipient countries most exposed along one dimension—financial integration, U.S. trade share, currency flexibility—tend also to be exposed along the others, so the conduits are best understood as overlapping rather than additively separable. This motivates the heterogeneity analysis of Section 4.9, which characterises directly how these same exposure dimensions govern the cross-country size of the spillovers.

Figure 10: Channel structural β paths



Note: Second-stage panel LP impulse responses for each channel mediator. Financial and FX channels: daily local projection per country, country-specific peak horizon selected as $h_i^* = \arg \max_h |t_i^h|$, GK-aggregated to monthly frequency. Trade channel: mediator is $USTS_{i,t} \times s_t$, where USTS is bilateral U.S. trade as a percent of country GDP. All specifications include country fixed effects. Standard errors: two-way clustered by country and time. Shaded areas denote 68% and 90% confidence bands.

4.8 Comparison with a demand-side bond market shock

As an additional robustness check, and to contextualize the magnitude and profile of my supply-shock estimates, I re-estimate all specifications replacing the Treasury supply shock with the LSAP (large-scale asset purchase) component of U.S. monetary policy surprises extracted by Swanson (2021). The LSAP shock is a demand-side bond market shock: it shifts the demand schedule for long-duration Treasuries rather than the supply schedule, and thus provides a natural foil for my main results.

This comparison is motivated by an established literature documenting that U.S. monetary policy surprises and in particular unconventional tools such as LSAPs, generate powerful global spillovers. Miranda-Agrippino and Rey (2020) show that a single global factor driven by U.S. monetary policy accounts for a substantial share of variation in risky asset prices worldwide. The international dimension is well documented for major economies: U.S. LSAP announcements compressed sovereign yields in the euro area, the United Kingdom, and Japan, and generated capital flows to emerging markets (e.g., Swanson, 2021). Carrière-Swallow et al. (2025) document that Japan monetary conditions contribute to global financial cycle, with heterogeneity that maps closely to investor exposure. This prior evidence on demand-side spillovers provides a natural benchmark: if my supply-shock estimates are in the same ballpark in terms of transmission elasticities, it would

suggest that global bond markets respond systematically to any repricing of U.S. long rates, regardless of whether the impulse originates on the supply or demand side.

An important conceptual point for this comparison is that both the Treasury supply shock and the LSAP demand shock are identified as *market responses to unanticipated surprises* in narrow event windows. In both cases, the relevant magnitude is the size of the market surprise, not the absolute scale of the underlying government intervention. The fact that the Treasury is the sole issuer of U.S. government debt (and hence constitutes 100 percent of the supply side) while the Federal Reserve is only one of many participants on the demand side does not affect the comparison, because what matters for the yield impact is not the market share of the actor but the extent to which the announcement deviates from market expectations.⁸ The near-identical transmission profiles I document below are consistent with this interpretation.

Turning to the results, the LSAP and supply shocks move yields in opposite directions—the LSAP compresses yields whereas the debt expansion shock raises them—but after sign-adjustment the transmission profiles for both domestic macro aggregates and international spillovers are strikingly similar in shape (Figure 11). The sign-adjusted responses of U.S. IP, 10-year yields, and foreign IP align closely across the two shocks, for both advanced economies (including the euro area and Japan) and emerging markets. This symmetry suggests that the mechanisms driving international transmission are a feature of U.S. bond market conditions broadly construed—the repricing of long-term rates and the attendant portfolio rebalancing—rather than a specificity of the supply-side channel. Coefficient magnitudes differ somewhat, partly because the LSAP shock is estimated over a somewhat shorter sample (1991–2019), but the qualitative patterns and the relative size of domestic versus international spillovers are consistent across both shocks. This robustness is reassuring: it indicates that my supply-shock results are not an artifact of the identification strategy but reflect a systematic feature of how U.S. bond market shocks propagate globally.

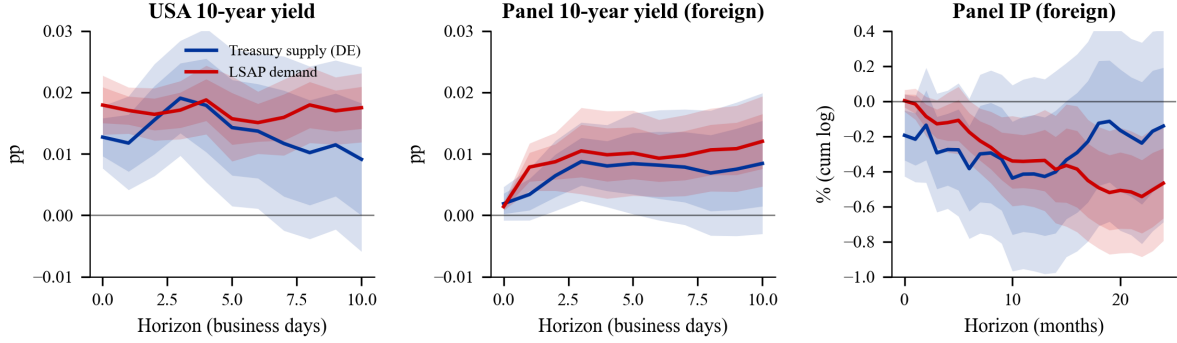
4.9 Heterogeneity in spillovers

I now turn to cross-country heterogeneity, asking which country characteristics amplify or attenuate the transmission of U.S. bond supply shocks. I organize the results around two distinct dimensions: financial market integration (which governs exposure to the financial channel) and macroeconomic fundamentals (which I further separate into *macro strength*—the ability to weather the shock through policy—and *borrowing capacity*—the extent to which the shock tightens the country’s own external financing conditions).

Financial market integration. The financial channel operates most powerfully through countries that are deeply integrated with global capital markets. Two empirical proxies capture this directly: the share of sovereign debt held by foreign investors, which measures the intensity of

⁸To draw an analogy: a surprise 25 bp rate hike by the Fed moves markets not because the Fed is the only participant in the federal funds market, but because the announcement was not anticipated. The same logic applies to Treasury supply surprises: what moves yields is the unanticipated component, regardless of the Treasury’s market share.

Figure 11: Responses to Treasury supply shock and LSAP demand shock



Note: Comparison of responses to the Treasury supply shock and the LSAP demand shock (sign-adjusted): 10-year yield (top) and industrial production (bottom), U.S. and panel. Shaded bands denote 68% and 90% confidence intervals. Source: Swanson (2021); Phillot (2025); author's calculations. Regression specifications kept the same and only shocks replaced across for LSAP (sample 1991-2019) or TSY (sample 1998-2020). Shaded areas denote 68% and 90% confidence bands. Standard errors: HAC for U.S. regressions; two-way clustered by country and time for panel regressions.

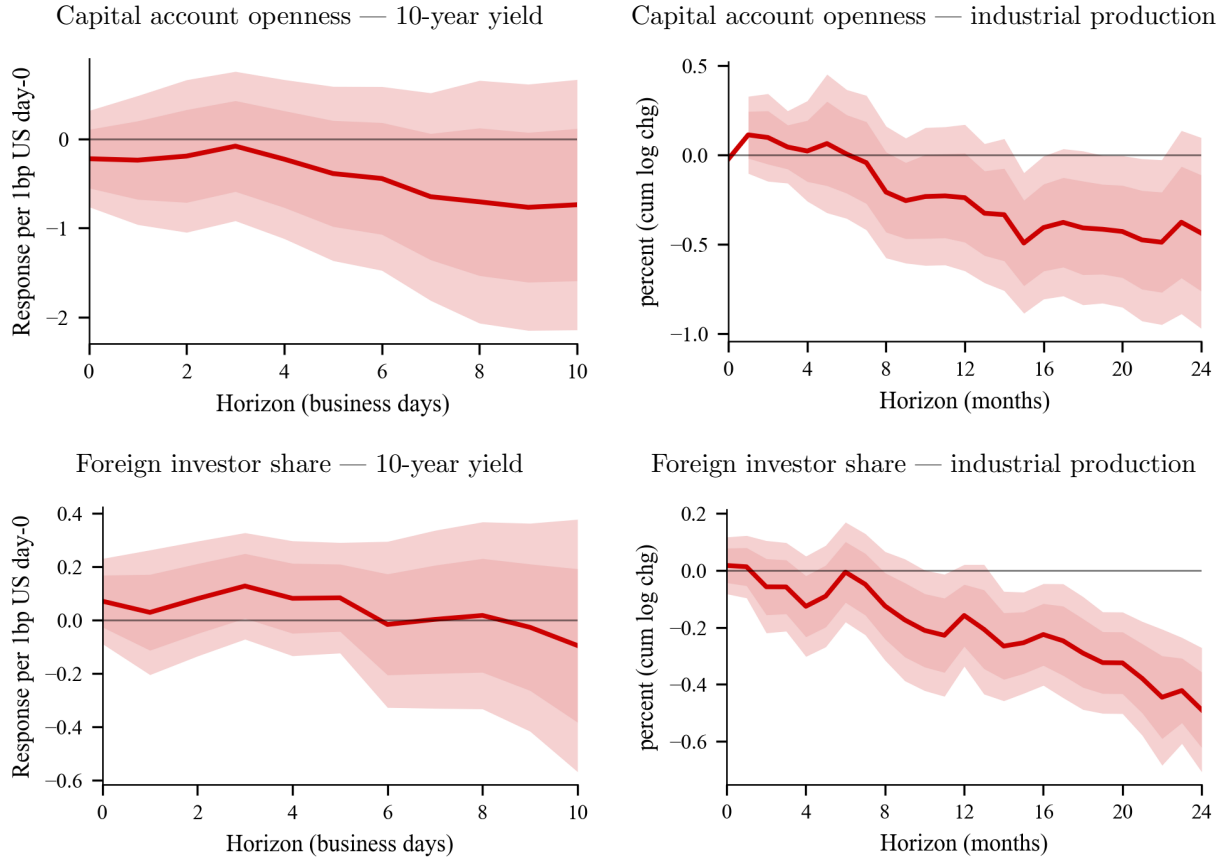
cross-border portfolio arbitrage, and capital account openness, which measures the legal and de facto freedom with which global investors can rebalance their portfolios across borders (Figure 12).

Countries with greater foreign investor participation experience significantly stronger spillovers particularly to industrial production. Moving from the 25th to the 75th percentile of the foreign investor share distribution deepens the industrial-production contraction by approximately 0.5 percentage points at a 6–12 month horizon. The differential yield response is of the expected positive sign but more modestly estimated over the daily horizon. The mechanism is a portfolio balance channel: when foreign investors hold a larger share of domestic sovereign bonds, a global repricing of safe assets triggers a reallocation of their portfolios that tightens local financial conditions for households, firms, and the sovereign itself. Economies with greater capital account openness exhibit a similar amplification pattern, consistent with this financial-integration interpretation. Taken together, these results confirm that countries better connected to international capital markets through financial linkages face the largest exposure to U.S. bond supply shocks.

U.S. interest rate beta. I also consider each country's sensitivity to U.S. interest rate conditions: the *U.S. interest rate beta*, defined as the 31-day rolling correlation between the country's 10-year sovereign yield and the VIX index. Countries with a high beta have yields that co-move more strongly with global risk conditions and, by extension, with U.S. long-term rates (Figure 13).

Splitting countries into high- and low-beta groups reveals that high-beta countries experience significantly stronger spillovers to 10-year sovereign yields following a U.S. debt expansion shock. This is intuitive: countries whose long-term rates are more sensitive to global risk repricing absorb a larger share of any shock to U.S. yields. However, the differential effect on industrial production across high- and low-beta groups is economically small and statistically insignificant, indicating that the stronger financial transmission does not necessarily translate into a differential real activity

Figure 12: Differential spillovers by financial market integration



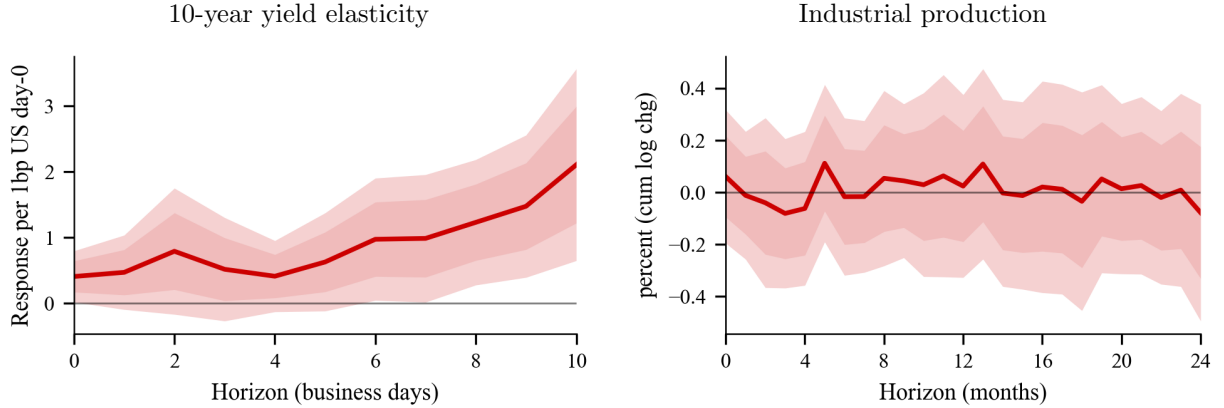
Note: Differential spillovers by financial market integration (IQR-scaled, 75th vs. 25th percentile). Top row: capital account openness (Chinn–Ito index). Bottom row: foreign investor share in sovereign bond markets. Left column: 10-year sovereign yield. Right column: industrial production. Standard errors: two-way clustered by country and time for all panels.

impact. One interpretation is that the real economy response depends on a broader set of factors—domestic monetary policy offsets, trade exposure, fiscal buffers—that are not well captured by the beta of long-term yields alone.

Macroeconomic strength: fiscal space and current account. Among countries with comparable financial integration, the severity of real spillovers depends critically on the ability of domestic policy to lean against the shock. I capture this dimension with two indicators of macroeconomic strength: fiscal space (the government’s buffer to accommodate countercyclical fiscal policy without triggering debt sustainability concerns) and the current account balance as a share of GDP (a summary of the economy’s reliance on external financing) (Figure 14).

Economies with larger fiscal space experience meaningfully smaller yield spillovers; the corresponding output-contraction differential is of the expected sign but more weakly estimated. The logic is direct: when a government has room to run temporary deficits, it can partially offset

Figure 13: Differential spillovers by U.S. interest rate beta (high vs. low)



Note: Differential spillovers for high- vs. low-beta countries (IQR-scaled). Beta is measured as the 31-day rolling correlation between the country’s 10-year sovereign yield and the VIX index. Left: 10-year sovereign yield. Right: industrial production. Standard errors: two-way clustered by country and time.

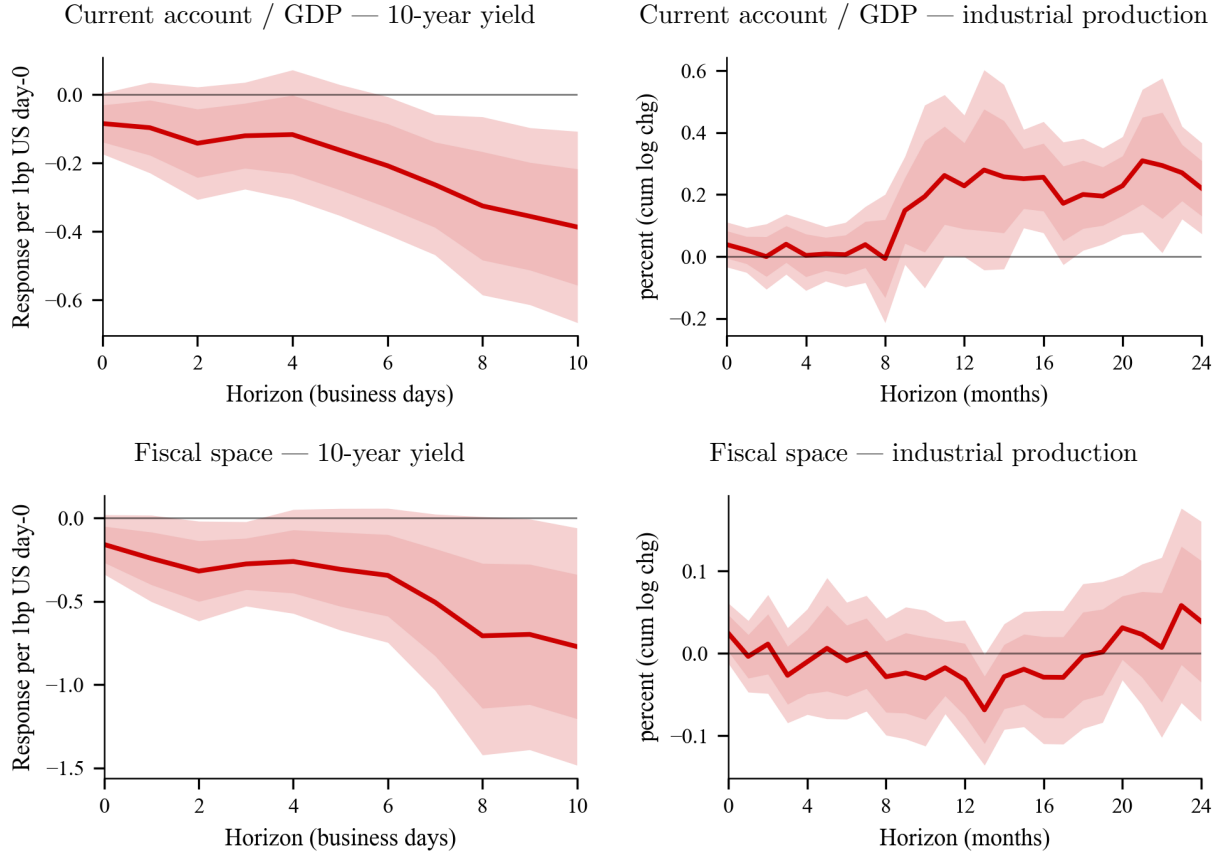
the contractionary impulse of higher borrowing costs through automatic stabilizers or active fiscal easing, dampening the transmission to real activity. Similarly, economies with stronger current account positions are less exposed: they do not depend on continuing external financing to sustain domestic demand, and a tightening in global financial conditions therefore represents a smaller disruption to their funding situation. The heterogeneity in fundamental resilience is economically sizeable—of similar order of magnitude to the amplification from financial openness—indicating that the same U.S. shock can produce very different macroeconomic outcomes depending on the recipient country’s policy buffers.

Higher debt-at-risk is associated with amplified spillovers to both yields and industrial production. The mechanism is a debt-sustainability amplifier: a global repricing of safe assets is transmitted more forcefully into the borrowing costs of fiscally fragile sovereigns, raising the perceived risk of holding their bonds and compressing their ability to refinance at reasonable cost. The interaction between fiscal fragility and external financial tightening thus deepens the real contraction in countries with the least fiscal headroom.

Exchange rate regime. I further examine whether the de facto exchange rate regime shapes the transmission of U.S. bond supply shocks. Using the IMF’s AREAER classification, I distinguish between countries with floating or freely floating exchange rates and those with more managed regimes (Figure 16).

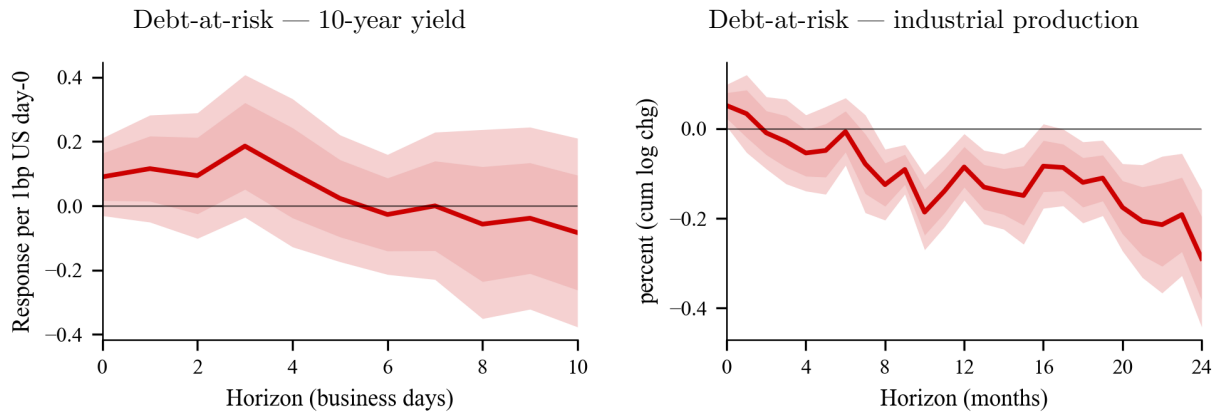
Countries with floating exchange rates exhibit marginally stronger spillovers to industrial production, consistent with the exchange rate channel documented in Section 4.7: a flexible exchange rate allows the U.S. dollar appreciation induced by the supply shock to pass through into a depreciation of the domestic currency, which tightens financial conditions through balance-sheet effects for dollar-indebted borrowers and compresses competitiveness for import-competing sectors. At the

Figure 14: Differential spillovers by macroeconomic strength



Note: Differential spillovers by macroeconomic strength (IQR-scaled, 75th vs. 25th percentile). Top row: current account balance-to-GDP. Bottom row: fiscal space. Left column: 10-year yield. Right column: industrial production. Standard errors: two-way clustered by country and time.

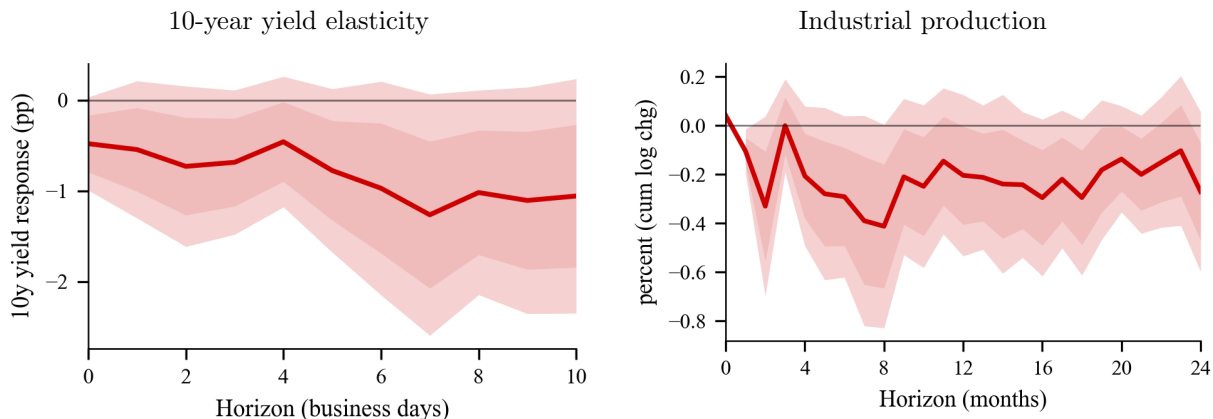
Figure 15: Differential spillovers by borrowing capacity



Note: Differential spillovers by borrowing capacity, measured by debt-at-risk (IQR-scaled, 75th vs. 25th percentile). Left: 10-year yield. Right: industrial production. Standard errors: two-way clustered by country and time.

same time, floating-rate countries display marginally *weaker* spillovers to 10-year sovereign yields. One interpretation is that a flexible exchange rate serves as a partial shock absorber for the bond market—the currency adjustment reduces the required repricing of domestic long-term rates—while amplifying the real-economy impact through the balance-sheet and competitiveness channels. The differences across regimes are modest in magnitude, however, suggesting that the exchange rate regime is a secondary moderator compared with financial integration and fiscal fundamentals.

Figure 16: Differential spillovers by exchange rate regime



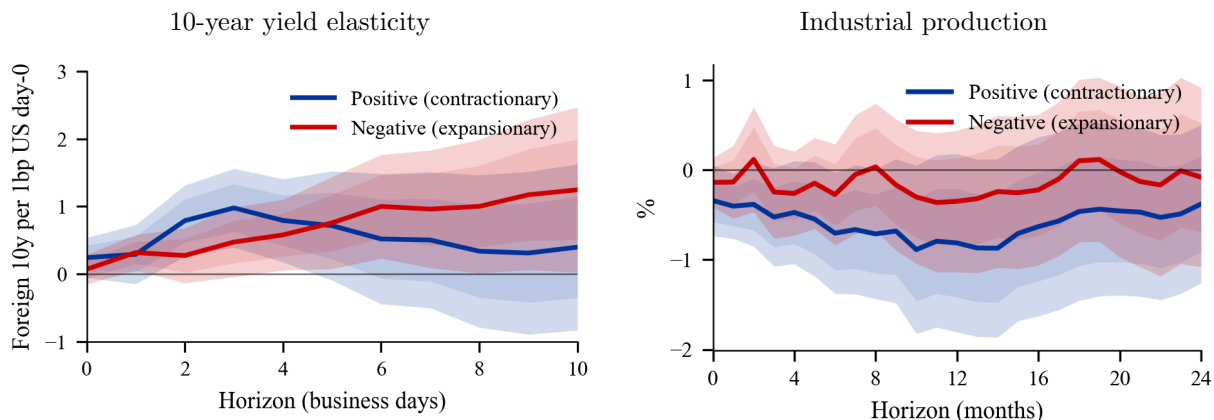
Note: Differential spillovers for floating/freely floating vs. managed exchange rate regimes (AREAER de facto classification). Left: 10-year sovereign yield. Right: industrial production. Standard errors: two-way clustered by country and time.

Asymmetry: positive versus negative shocks. Finally, I investigate whether the effects of contractionary (positive, rate-raising) and expansionary (negative, rate-lowering) Treasury supply shocks are symmetric (Figure 17). I split the shock series into positive and negative realizations and re-estimate the baseline specifications separately for each subsample.

Positive (contractionary) shocks that raise interest rates produce more pronounced effects on both U.S. and foreign yields and on industrial production than negative (expansionary) shocks of the same absolute magnitude. The asymmetry is particularly evident in the yield response at short horizons and in the IP response at medium horizons, where the contractionary shock generates responses noticeably larger than those of an expansionary shock of the same absolute magnitude. This finding is consistent with a well-established result in the monetary policy literature. Tenreiro and Thwaites (2016) find that contractionary monetary policy shocks have more powerful real effects than expansionary shocks, a result confirmed in cross-country settings by Angrist et al. (2018) and others. The theoretical mechanism is typically linked to downward nominal rigidities, credit constraints that bind more tightly in contractions, and the asymmetric nature of financial frictions: a tightening in financial conditions compresses investment and hiring through binding borrowing constraints, whereas an easing may not symmetrically relax those constraints if firms are already unconstrained. My finding that Treasury supply shocks display a similar asymmetry suggests that

the underlying transmission mechanism—operating through long-term rates, borrowing costs, and portfolio rebalancing—shares the same nonlinear features documented for conventional monetary policy.

Figure 17: Asymmetric effects: positive (contractionary) vs. negative (expansionary) shocks



Note: Impulse responses estimated separately for positive (contractionary, rate-raising) and negative (expansionary, rate-lowering) debt expansion shocks. Left: 10-year sovereign yield. Right: industrial production. Shaded bands denote 90% confidence intervals. Standard errors: two-way clustered by country and time.

5 Conclusion

This paper documents that unexpected changes in U.S. Treasury issuance constitute an important and previously underemphasized driver of global financial conditions and international real activity. Using high-frequency identification around auction announcements, I find that debt expansion shocks rapidly and substantially transmit to foreign sovereign bond yields with a sizeable elasticity, on the order of 0.7. Additionally, they generate meaningful contractions in foreign industrial production at medium horizons, with spillovers comparable in size to—but more persistent than—the domestic U.S. impact.

Three findings stand out for policy. First, the large daily spillover elasticity implies that changes in U.S. borrowing costs feed quickly into the borrowing conditions faced by foreign governments, amplifying the global financial tightening associated with increases in U.S. public debt issuance. Second, the transmission of real spillovers through the financial and trade channels, comparable in size to the domestic effect, suggests that standard models that abstract from cross-border financial linkages substantially underestimate the global cost of U.S. fiscal expansions. Third, the strong heterogeneity across countries implies that emerging markets and countries with high foreign investor participation, limited fiscal space, or poor market access are disproportionately exposed to shocks emanating from the U.S. bond market.

Appendices

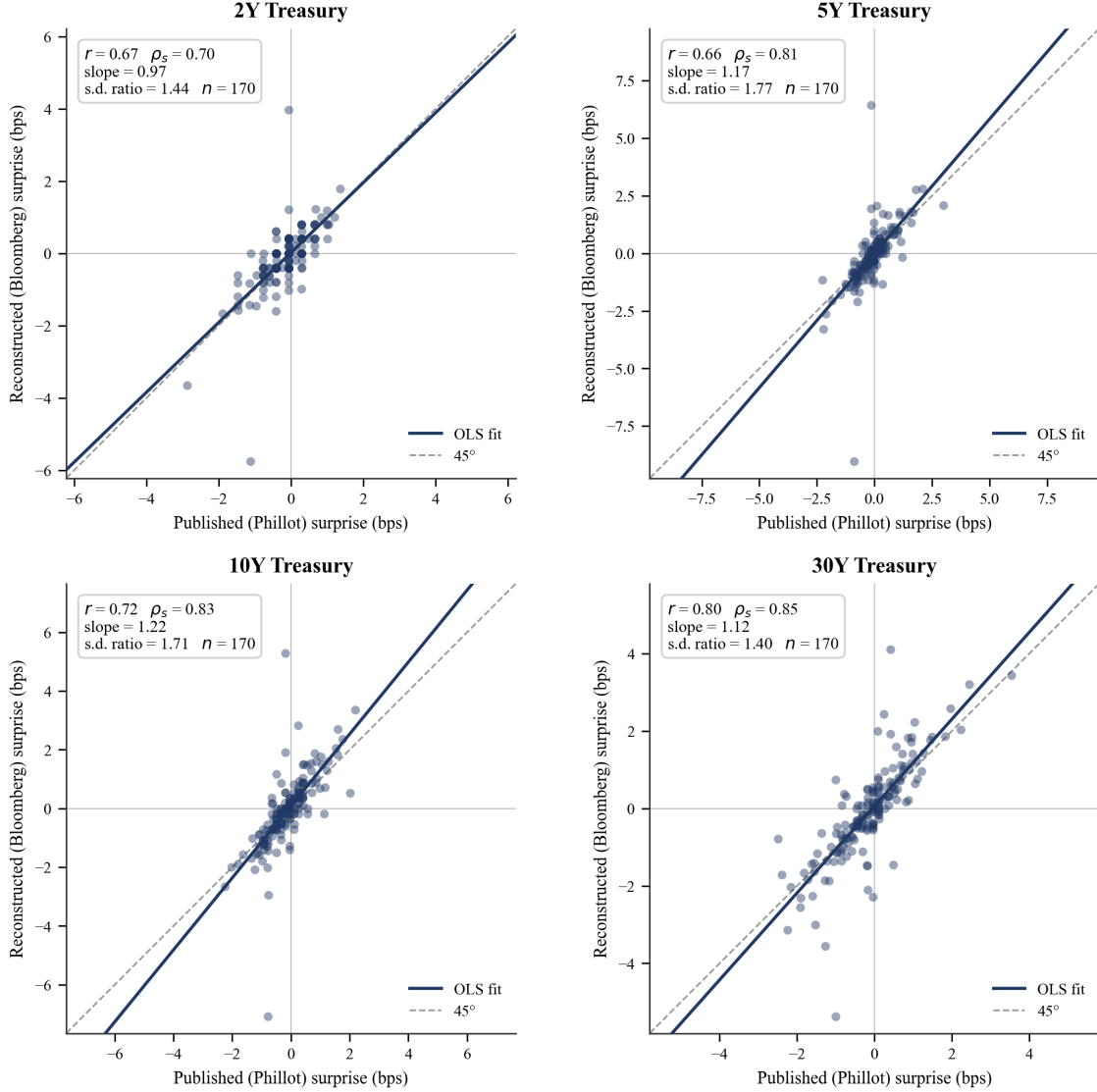
A Merging and Extending the Shock Series

The baseline shocks combine two sources identified with the same Phillot (2025) strategy. The first is the public Phillot (2025) series of within-window Treasury futures surprises, available through January 2020. The second is an extension I construct from intraday Bloomberg tick data for the 2-, 5-, 10-, and 30-year Treasury futures, taking the price change in the 30-minute window bracketing each scheduled auction announcement and flipping the sign so that a positive value denotes a contractionary supply surprise, exactly as in Phillot (2025). The merge proceeds in three steps. (i) *Overlap validation*: on the window where both series exist—the published series runs 1998–2020 and my reconstruction 2013–2026, so the two genuinely overlap over 2013–2020—I find close agreement. Within that window the two procedures flag essentially the same auction dates: the rolling share of matched event-days is around 90% throughout 2014–2019, and on the 170 matched dates the maturity-level surprises are strongly positively correlated with the published series (Pearson 0.66–0.80, Spearman 0.70–0.85), with OLS slopes close to one (0.97–1.22); see Figure 18. My reconstruction is somewhat noisier than Phillot’s cleaned futures-based surprises (standard-deviation ratio 1.4–1.8), as expected when surprises are read off Bloomberg yield ticks rather than from a curated futures dataset. I rescale my series to the published units before splicing. Crucially, residual classical measurement error of this kind is innocuous for my estimand: as shown in Section 3.2, it attenuates the first stage and the reduced form proportionally and cancels in the IV spillover ratio. (ii) *Splicing*: I append the post-2020 reconstructed surprises to the published series at each maturity, taking care that the rename and re-index operations are idempotent so that no auction date is duplicated and post-2020 observations are not silently dropped by index misalignment. (iii) *Re-decomposition*: I re-run the Bi et al. (2026) principal-component decomposition on the spliced cross-maturity surprise matrix, so that the debt-expansion (level) and maturity-extension (slope) factors and their loadings are estimated on the full extended sample. The resulting debt-expansion and maturity-extension series coincide with the original ones over the pre-2020 period and extend coverage through the post-COVID period.

A.1 Validation: within-window variation, true vs. placebo windows

As a check that the extended/spliced windows carry genuine supply information rather than noise, I compare the dispersion of the within-window futures yield change on true announcement windows against placebo windows—a fixed 11:00 ET benchmark on randomly selected non-announcement business days, following Känzig (2021) and Phillot (2025). Figure 19 plots the rolling standard deviation of the within-window change for the two sets of windows over the sample. The true-announcement windows are visibly and persistently more dispersed: a variance-ratio test rejects equality of variances, and a two-sample Kolmogorov–Smirnov test on the absolute within-window changes rejects equality of distributions, with the true windows exhibiting thicker tails and more

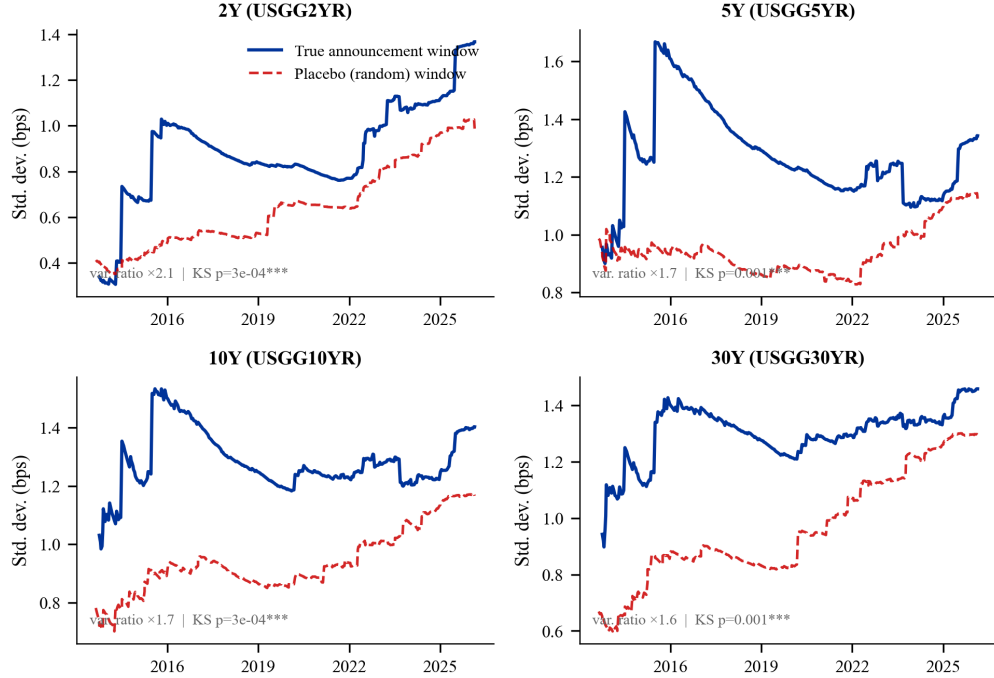
Figure 18: Reconstructed vs. published surprises on the overlap window



Note: Bloomberg-reconstructed within-window Treasury surprises against the published Phillot (2025) surprises on the 170 matched auction days in the 2013–2020 overlap window, by maturity (basis points). Pearson correlations range from 0.66 (5Y) to 0.80 (30Y); Spearman rank correlations from 0.70 to 0.85; OLS slopes from 0.97 to 1.22. The reconstructed series is somewhat more dispersed (standard-deviation ratio 1.4–1.8), reflecting greater tick-level measurement noise. The match rate is low as a share of the calendar union only because the two series cover different eras; within the genuine 2013–2020 overlap the rolling date-match is around 90% (Section 2.1).

frequent large moves. This mirrors the placebo evidence in Bi et al. (2026) and confirms that the windows in the extended sample remain informative.

Figure 19: Rolling within-window variation: announcement vs. placebo windows



Note: Rolling standard deviation of the within-window Treasury futures yield change (basis points) on true auction-announcement windows versus placebo windows (a fixed 11:00 ET benchmark on non-announcement business days). The reported variance ratio and two-sample Kolmogorov–Smirnov test compare the full distributions of absolute within-window changes.

A.2 Robustness: results on the public (pre-2020) sample

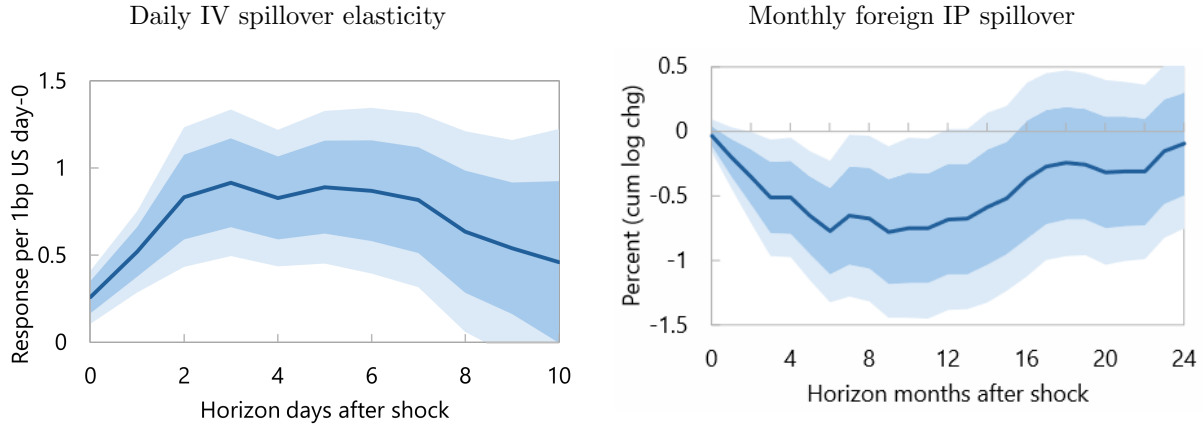
Because the post-2020 extension is my own reconstruction, I verify that none of the paper’s conclusions depend on it. Figure 20 re-estimates the two headline objects—the daily IV spillover elasticity and the monthly foreign-IP spillover—on the public Phillot (2025) auction sample alone (through January 2020), using the identical specification. Both are quantitatively close to the full-sample estimates of the main text, confirming that the extension sharpens precision and enables the post-COVID comparison of Section 4.4 without altering the substantive findings.

B Maturity Extension Shock: Supplementary Results

C Granger Causality Pre-Tests

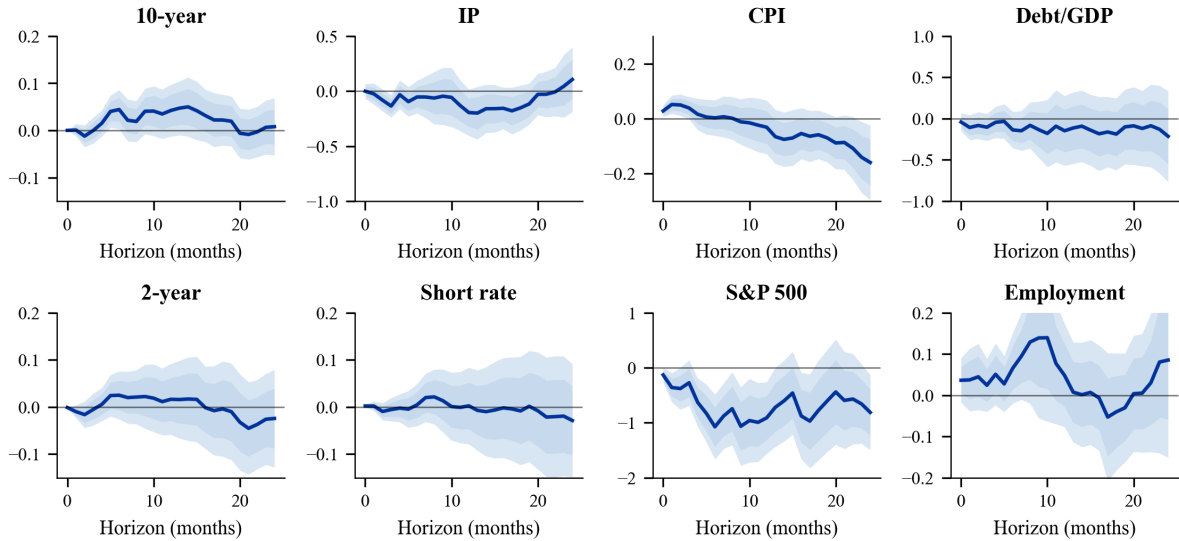
Table 2 reports p -values for the null hypothesis that each candidate control variable does not Granger-cause the debt expansion shock. Variables for which the null is rejected at the 10 percent

Figure 20: Headline results on the public (pre-2020) sample



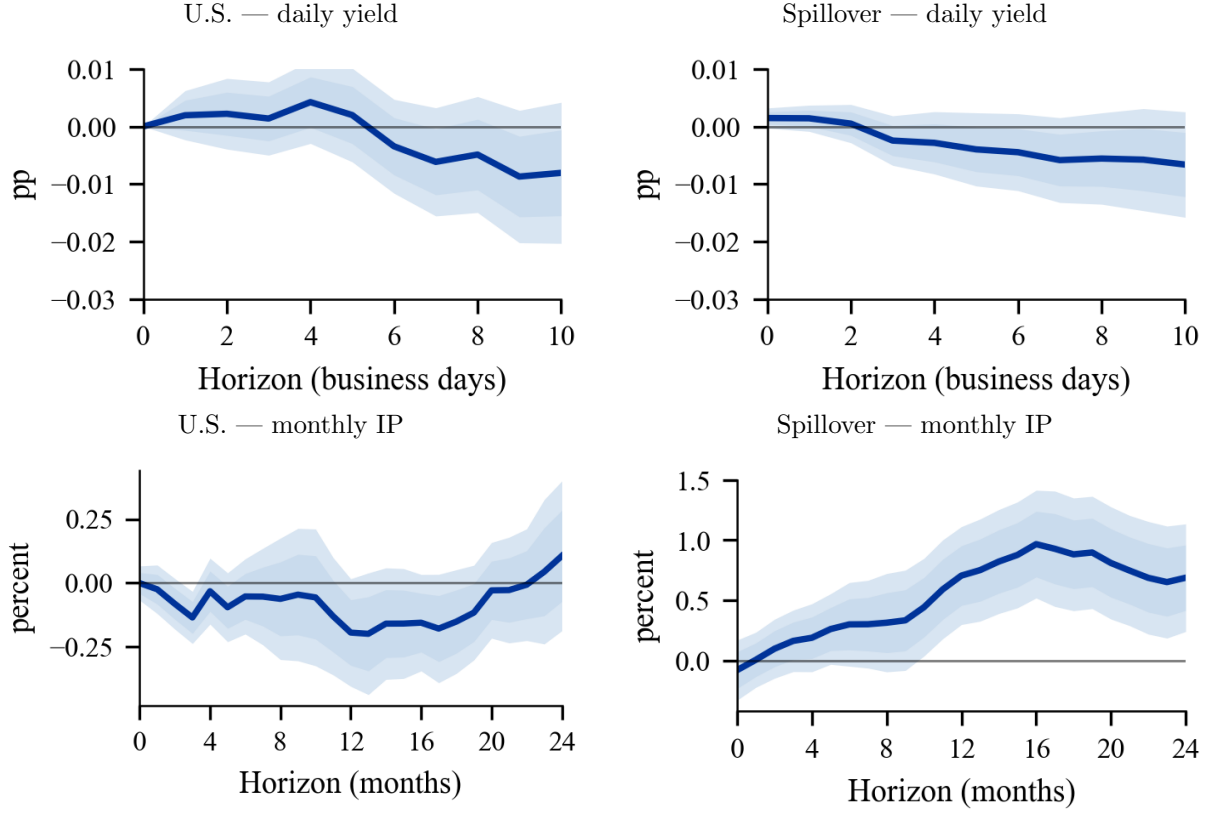
Note: Key results re-estimated on the public Phillot (2025) sample (auctions through January 2020), identical specification. Left: IV-normalized daily spillover elasticity (cf. Figure 3). Right: monthly foreign panel IP response (cf. Figure 8). Bands as in the corresponding main-text figures.

Figure 21: U.S. monthly response to maturity extension shock



Note: U.S. monthly impulse responses to the maturity extension shock (analogous to Figure 6). Standard errors: HAC (Newey–West, bandwidth $h + 1$).

Figure 22: U.S. and foreign panel industrial production response to maturity extension shock



Note: Monthly impulse responses of U.S. and foreign panel industrial production to the maturity extension shock (analogous to Figure 8). Standard errors: HAC for U.S.; two-way clustered by country and time for panel.

level are included as controls (contemporaneous value and two lags) in all regressions.

Table 2: Granger causality

variable (daily)	Debt expansion p-value	Maturity extension p-value
first difference of 10-year yield US	0.47	0.14
first difference of 10-year yield Germany	0.40	0.13
first difference of 10-year yield Japan	0.14	0.28
first difference of 2 year term premium US	0.86	0.07
first difference of 5 year term premium US	0.85	0.08
first difference of 10 year term premium US	0.88	0.10
first difference of 10-year break-even inflation US	0.26	0.44
VIX	0.51	0.40
log first difference of NASDAQ	0.05	0.69
log first difference of S&P 500	0.40	0.59
log first difference of Oil price	0.48	0.11

Note: Granger causality pre-tests: p -values for the null hypothesis that each variable does not Granger-cause the shock. Daily data; lag length selected by minimum p -value up to 2 lags.

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